

Kingdom of Saudi Arabia
Ministry of Education
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Software Engineering

Term 1 – 2024/2025

Software Design Specifications For Elmuna App



Version 1.0

CS Year 4, G1

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Date of preparation

Monday, Nov 20, 2025

Software Design Specifications for Ellmuna App

This Software Design Specification was prepared and provided as a deliverable for Software Engineering, CS411, Term 1 and it will be used by [name of end user].

This document is based in part on the IEEE Recommended Practice for Software Design Descriptions.

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Revision History

Name	Date	Reason For Changes	Version
All members	Oct. 30, 2025	Prepared initial version	0.1
All members	Nov. 1, 2025	Updated section 1	0.2
All members	Nov. 3, 2025	Updated section 2	0.3
All members	Nov. 5, 2025	Updated section 2,3	0.4
All members	Nov. 10, 2025	Updated section 4,5	0.5
All members	Nov. 15, 2025	Updated section 6,7	0.6
All members	Nov. 20, 2025	Complete review - Final version	1.0

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1. Introduction

overview

Software design is a process by which the software requirements are translated into a representation of software components, interfaces, and data necessary for the implementation phase. This SDD describes how the requirements, as set out in the Ellmuna SRS, will be transformed into a complete software design. It outlines the architecture, major components, data structures, and interfaces guiding the implementation phase.

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Ellmuna is an academic application designed for Saudi Arabian students. It allows students to calculate the weighted score for high school students and GPA for university students and navigate universities in Saudi Arabia by degree levels. This SDD describes the internal architecture of these features to offer the functionalities described here accurately, under usability, and for efficient performance.

This document follows the IEEE guidelines for software design specifications and outlines both the preliminary high-level architecture of the system and a detailed design for each module. As such, this will provide a basis for developers, testers, and supervisors during system development to ensure that what is ultimately produced is in line with the functionality required.

1.1 Purpose

This document aims to outline and describe the software design for the Ellmuna Application. This document takes the functional and non-functional requirements identified in the SRS and turns them into a structured design to guide the development process. It serves as a blueprint for the programmers to ensure that the system's implementation aligns with the intended architecture, logic, and functionality.

1.2 Scope

The Ellmuna Application is intended to help high school and university students in managing their academic information effectively. The key features of the application are:

- Weighted Score Calculator for high school students.
- University students' GPA calculator.
- Saudi Universities browser to explore programs.
- Settings module for customization/theme/welcome screen.

Usability, accuracy, and a simple interface for users are key concerns of the system, while the administrative functions are limited to managing data related to universities and programs.

1.3 Definitions, Acronyms, and Abbreviations

Acronyms are listed in the table below:

Acronyms	Definition	Stands for
SDD	A document that outlines the system's architecture, components, data structures, and design details necessary for implementation.	Software Design Description
SRS	A formal specification containing all functional and non-functional requirements of the system.	Software Requirements Specification

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UI/UX	UI refers to visual and interactive elements; UX describes the overall experience and usability of the system.	User Interface / User Experience
API	A set of communication rules that enable interaction between software components or external services.	Application Programming Interface
GPA	A numerical indicator of academic performance used in universities.	Grade Point Average
PhD	individuals with doctoral-level expertise who contribute to academic or technical review.	Doctor of Philosophy
GUI	Interface elements that allow users to interact visually with the system.	Graphical User Interface
ER	A data modeling approach used to represent system entities, attributes, and their relationships, typically in database design.	Entity–Relationship
UML	A standardized modeling language used to visualize and document the structure and behavior of software systems.	Unified Modeling Language
DFD	A diagram showing how data moves through the system, including processes, data stores, inputs, and outputs.	Data Flow Diagram
PDL	A structured English-like language used to describe program logic and algorithmic steps before implementation.	Program Design Language (or Pseudo-code Description Language)

Table 1 Acronyms and their Meaning

1.4 References

- [1] "Software Requirement Specification (SRS) for Ellmuna App," Dammam, 2025.
- [2] IEEE, "IEEE Recommended Practice for Software Design Descriptions," New York, 1998.
- [3] E. Wong, "User Interface Design Guidelines: 10 Rules of Thumb," 2023. [Online]. Available: <https://www.interaction-design.org/literature/article/user-interface-design-guidelines-10-rules-of-thumb>.
- [4] IEEE, "IEEE Standard for Software Project Management Plans," The Institute of Electrical and Electronics Engineers (IEEE), New York, 1998.

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2. System overview

Ellmuna is a mobile app created to make academic tasks easier for students in Saudi Arabia. It brings together the most important educational tools in one place, so both high school and university students can check their grades, calculate their scores, and explore available majors across Saudi universities without needing multiple websites.

The idea behind Ellmuna is to keep everything accurate, simple, and easy to use. The system is organized in a way that separates the interface, the logic, and the data, which makes the app more stable and allows new features to be added smoothly in the future. Its main tools include calculating the high school weighted admission score, computing the university GPA, and browsing universities depending on degree level.

Ellmuna app also focuses a lot on clarity. It adds small guides and explanations across different pages to help students understand how calculations work—like grade scales, weighted score formulas, and how to read their results. Also, Ellmuna focus on provide a reliable academic space where students can find the information they need quickly, instead of searching through scattered online sources.

2.1 System Functionality

2.1.1 Weighted Score Calculation

Ellmuna provides high school students with an accurate calculator for determining their weighted admission score. Users can input:

- High school percentage
- Qiyas score
- Tahsily score

The system applies official Saudi formula weights and outputs a final score used for university admission.

2.1.2 GPA Calculation

University students can calculate their updated GPA by entering both previous academic data and current semester courses. The page is divided into two main sections:

Previous Academic Record:

- Previous GPA input field
- Total previous credit hours input field

Current Semester:

- A list of courses where the user can enter credit hours and select a grade for each course
- An Add Course button to insert additional course rows
- A delete icon to remove any course row

After entering all required information, the user can press the Calculate My GPA button to generate the result.

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2.1.3 Saudi Universities

This feature provides users with a categorized and searchable list of Saudi universities and their programs. It helps users explore available Bachelor's, Master's, and Doctorate degrees across different institutions. This feature is of Medium Priority, as it provides useful but non-critical academic information.

2.1.4 Settings

This feature allows users to customize their application preferences. Users can manage display options, such as enabling or disabling the welcome screen and switching between light and dark mode. This feature is of Medium Priority, as it enhances usability and comfort but is not required for core functions.

2.2 Design Approach

Ellmuna adopts a component-based architecture where each primary feature (Weighted Score, GPA, University Navigation) is implemented as an independent module. These modules communicate through a shared logic layer that handles calculations, validations, and data processing.

The system is structured into:

- Presentation Layer: Responsible for UI/UX and user interactions.
- Logic Layer: Implements all computational algorithms.
- Data Layer: Stores static data such as university lists, GPA mappings, and degree-level classifications.

This design ensures flexibility, maintainability, and high performance.

2.3 Background information

Saudi students often rely on manual calculations or third-party websites to obtain their weighted score or GPA, which may lead to inconsistent or inaccurate outcomes. Ellmuna consolidates these essential tasks into a single, reliable tool that follows official standards from the Ministry of Education and recognized Saudi universities.

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3. Design Considerations

Issues that will be addressed in this section include the assumptions and dependencies of the Ellmuna system, including the constraints that will be implemented.

3.1 Assumptions and Dependencies

The assumptions and dependencies of the system can be grouped into several categories, which are:

- Relating software or hardware.
- operating system.
- The characteristics of the end user.
- Possible and/or probable changes in functionality.

3.1.1 Related Software or Hardware

There are software and hardware dependencies in Ellmuna, such as the fact that a stable Internet connection is required in order to browse university data. However, GPA and weighted score calculations can function offline, since the calculations are processed locally on the device. The application depends on the availability of updated academic datasets stored in the cloud server. Moreover, Ellmuna depends on the following mobile device capabilities:

- Updated web view components for displaying content.
- Adequate storage for catching university lists.
- Cloud hosting services to retrieve university programs and metadata.

No external hardware is needed other than a common mobile device.

3.1.2 Operating System

The Ellmuna system is intended to run on iOS and Android mobile devices. The minimum supported versions are:

- ios 14 or newer.
- Android 10 (API 29) or newer This guarantees compatibility with modern UI components, secure connections, and proper performance across devices.

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3.1.3 End-User Characteristics

End-User	Age	Gender	Education Level	Computer Skills	Region
High School Student	15–20	Male/Female	High school level	Fair/Good	KSA
Bachelor Student	18–26	Male/Female	Undergraduate	Good/Very Good	KSA
Master/PhD Applicant	22+	Male/Female	Bachelor/Master	Good/Very Good	KSA
General User/Parent	25+	Male/Female	Variable	Fair/Good	KSA

Table 2 End-User Characteristics

3.1.4 Possible and/or Probable Changes in Functionality

The design of Ellmuna can be expanded in later versions by including more academic services such as:

- Direct integration with university APIs for real-time admission data.
- Adding a login system to store personal academic history.
- Supporting notifications for academic deadlines.

Since design and the engineering of requirements are interconnected, every change in system design might affect system requirements and vice versa.

3.2 General Constraints

While designing the Ellmuna system, the following are a few constraints that should be kept in mind:

- Hardware or software environment.
- Interface protocol requirements.
- Data repository and distribution requirements.
- Security and protection requirements.
- Performance requirements.
- Verification and validation requirements (Testing).

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3.2.1 Hardware or Software Environment

As explained earlier, Ellmuna can run on mobile operating systems like:

- Android 10+
- iOS 14+

The application pulls required university data from a cloud repository.

GPA and weighted score calculations are run locally without requiring any external servers.

3.2.2 Interface Requirements

Ellmuna offers an appealing and easy-to-use GUI. The interface has to adhere to the following:

- Clear input fields of scores and GPA values.
- Consistent error and pop-up messages. Unified font style and readable
- Organized layout for browsing Saudi universities.
- High-resolution images and logos.
- Responsive design for different mobile screen sizes.

3.2.3 Data Repository and Distribution Requirements

User-submitted data (scores, GPA inputs) are locally processed and not saved. University lists, majors, and programs are stored within a cloud database and retrieved when needed.

The repository must support:

- Fast data retrieval.
- Structured categories:(university → college → program).
- Periodic updating of academic information.

3.2.4 Security and Protection Requirements

Since Ellmuna does not require login or store personal accounts, security focuses on:

- Protecting communication between the app and server with HTTPS and TLS encryption.
- Ensuring that academic data sets retrieved from the cloud cannot be tampered with.
- Validation of user inputs - GPA, hours, scores against invalid or harmful data entry.
- Limiting the right to edit academic datasets solely to administrators.

3.2.5 Performance Requirements

- Immediate computation of GPA and weighted scores by the system.
- University lists should load in 1–3 seconds depending on network speed.
- The application should work seamlessly on low to mid-range devices.
- UI must respond promptly for every action.

3.2.6 Verification and Validation Requirements (Testing)

Verification and validation ensure that the Ellmuna system performs correctly by:

- Conducting the functional testing for GPA and weighted score accuracy.
- Conducting usability testing to ensure smooth navigation.
- • Verification that all academic information is correctly displayed.
- • Following test-driven development to minimize errors.
- Making sure that automated tests for future versions are established.

4. User Interface Design

4.1 Overview of User Interface

The *Ellmuna* application provides a clean and intuitive user interface designed to help users navigate the system smoothly. From the user's perspective, the system offers access to three primary features: calculating the weighted average for high school students, calculating university GPA (both term and cumulative), and browsing Saudi universities, colleges, and majors.

Each screen is designed interactively, allowing the user to enter the required data in structured input fields, click action buttons to perform calculations, and receive instant feedback messages—such as performance evaluation or result interpretation. A side navigation menu is also available, giving the user access to settings, welcome screen control, and dark mode options.

Overall, the interface ensures that the user can complete all expected features easily, with clear guidance, smooth transitions, and helpful visual feedback.

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4.2 Interface Design Rules

The *Ellmuna* interface design follows a set of principles to ensure clarity, usability, and consistency across the application. The main design rules include:

1. **Consistency:**
All screens follow a unified design pattern in colors, typography, spacing, and button style.
2. **Simplicity:**
Only essential elements are displayed to avoid overwhelming the user.
3. **Logical Flow:**
Each feature guides the user step-by-step from data input, to processing, to result display.
4. **Immediate Feedback:**
Users receive instant responses after performing actions such as calculations.
5. **Usability:**
Buttons, input fields, and icons are sized and positioned to ensure ease of interaction.
6. **Theme Support:**
The interface supports dark mode as part of enhancing the user experience.

4.3 Screen Images

1- Home Screen :

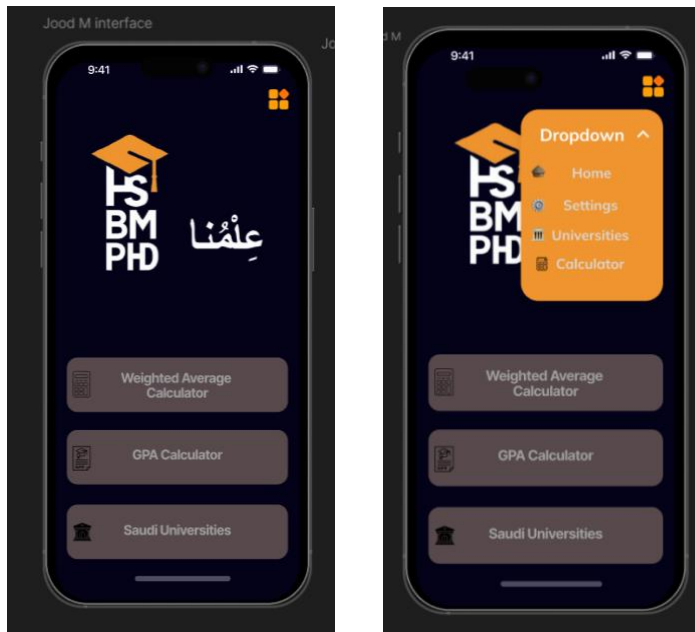


Figure 1 Home Screen and Dropdown Menu Interface

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2. When the user chose first option “Weighted Average Calculator” :

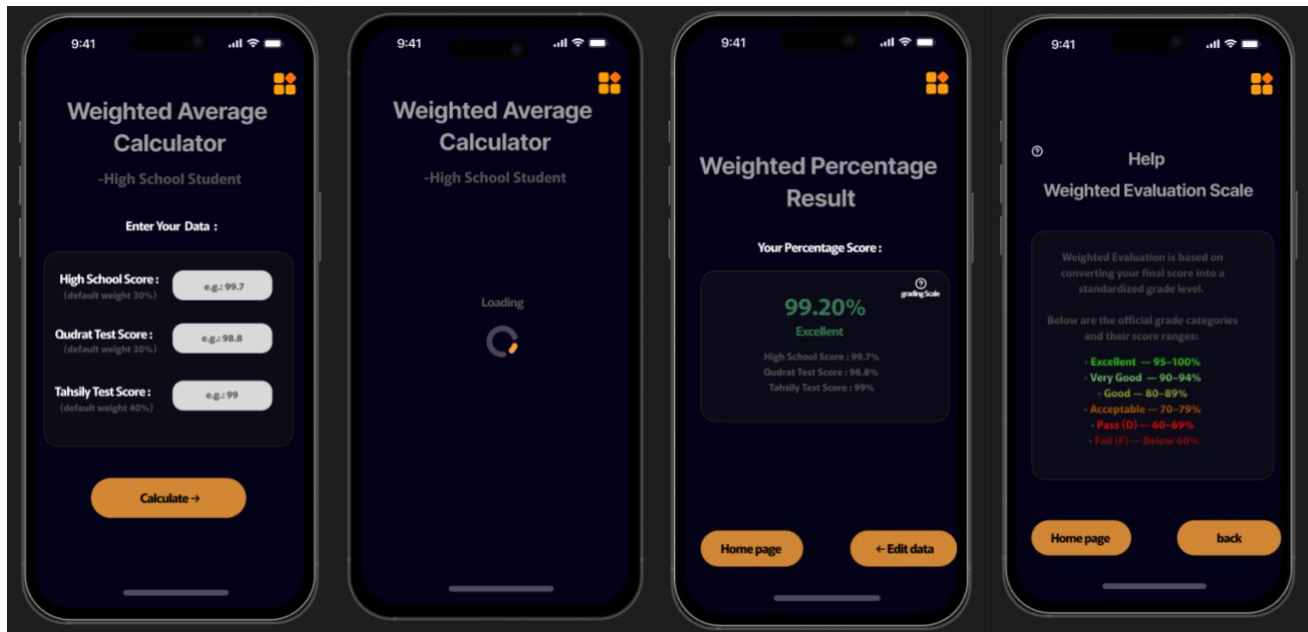


Figure 2 : Weighted Average Calculator workflow showing the input screen, loading screen, percentage result display, and the weighted evaluation scale help screen.

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3. When the user chose the second option “GPA Calculator” :

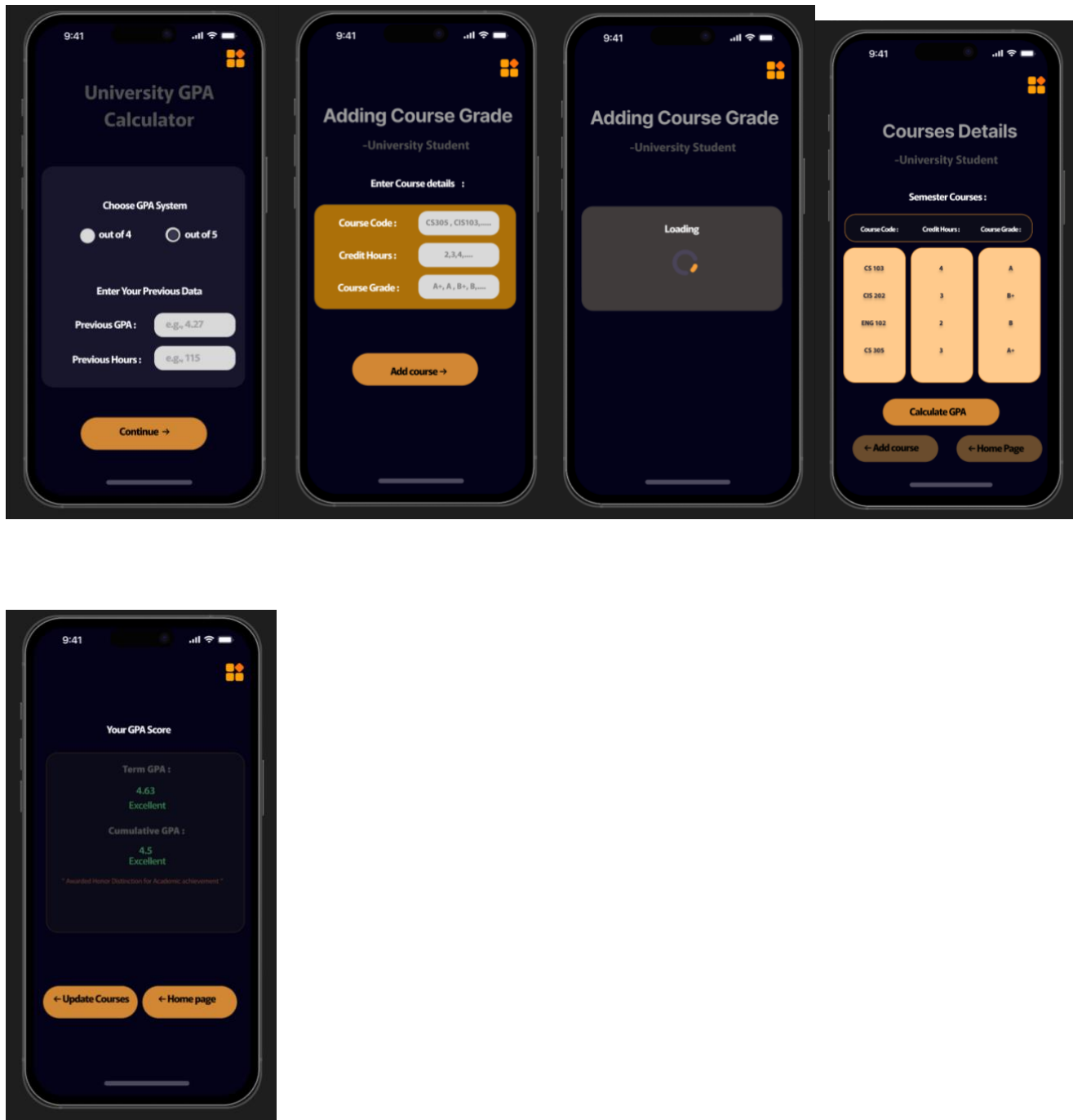


Figure 3 : **GPA Calculator Screens (User Flow)** This figure shows all the main screens displayed when the user selects the “GPA Calculator” option.

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4. When the user chose “Saudi University”:

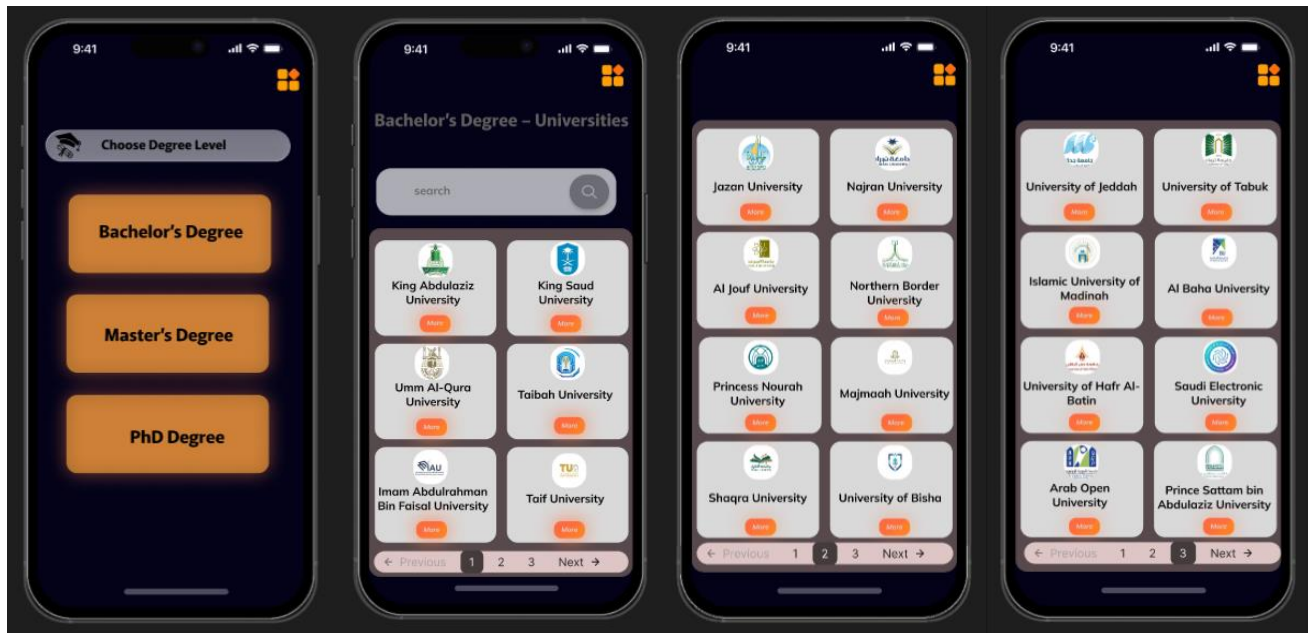


Figure 4 : User interface screens showing the process of selecting a degree level and browsing Saudi universities.

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- If the user chose “Bachelors Degree”:

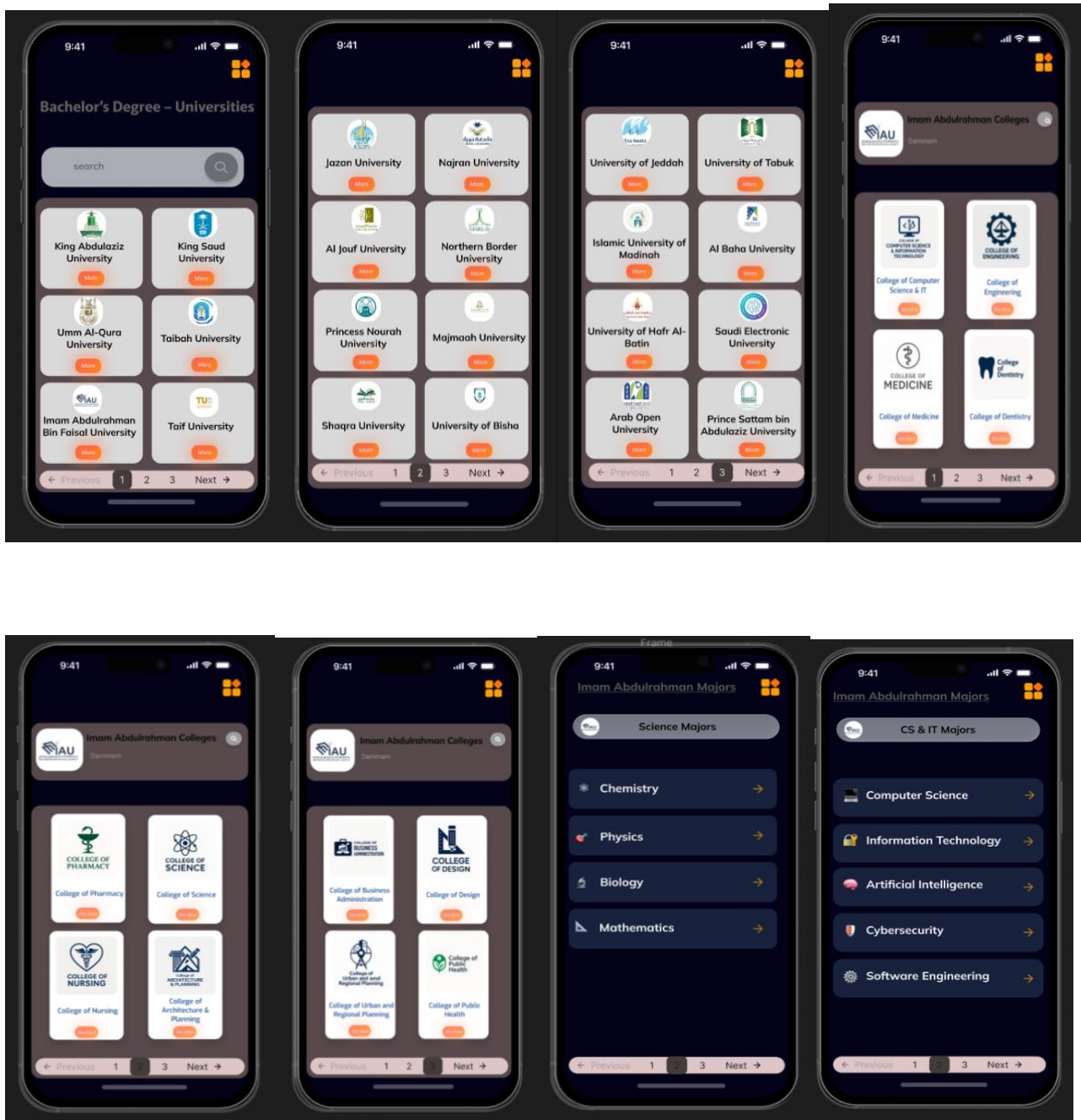


Figure 5 User interface screens showing the process of browsing Bachelor's Degree universities, viewing available colleges, and exploring majors within each college.

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- If the user chose “Master's degree”:

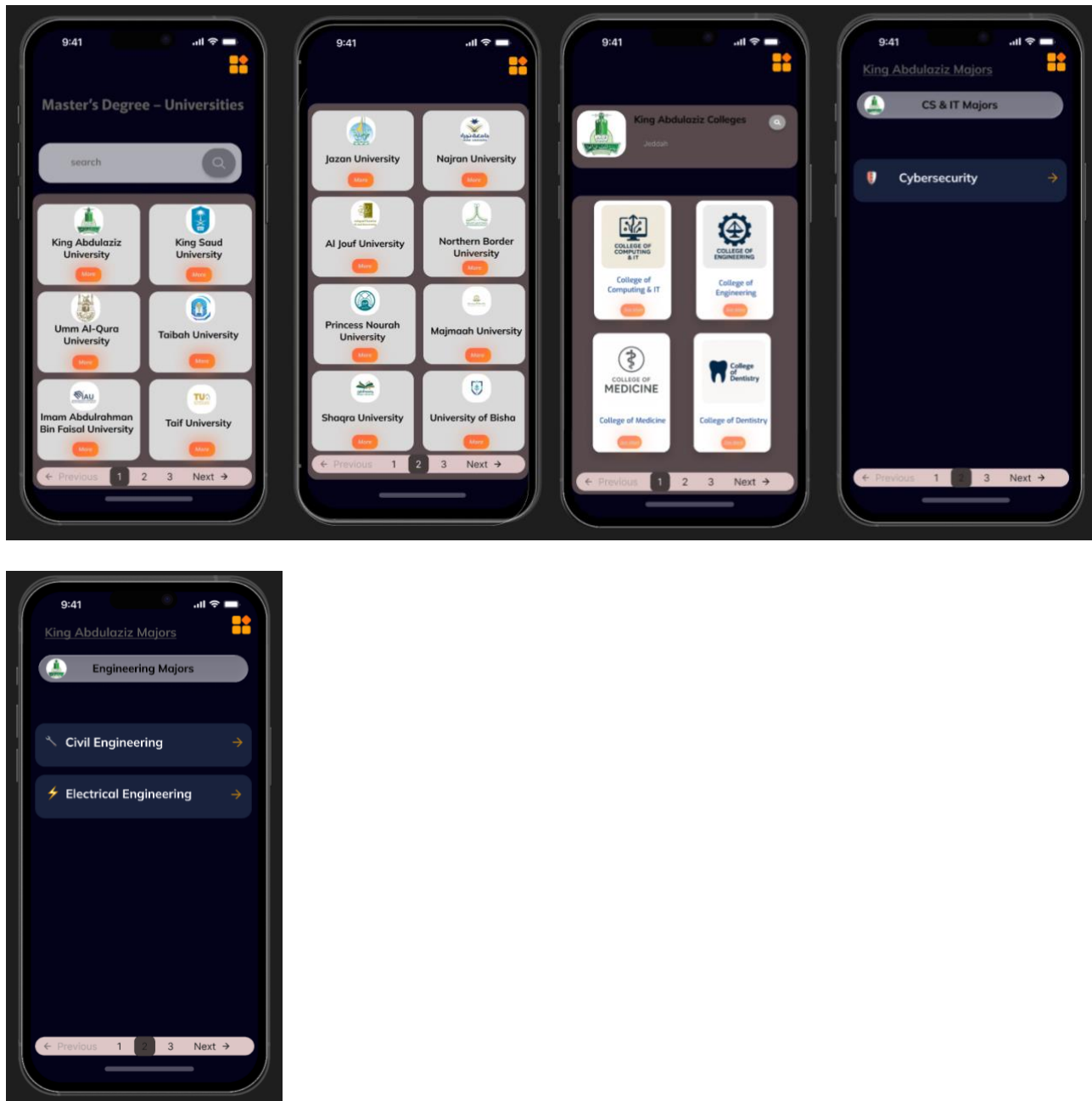


Figure 6 User interface screens for browsing Master's degree universities and navigating through colleges and majors in the selected university.

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- If the user chose “PhD Degree”:

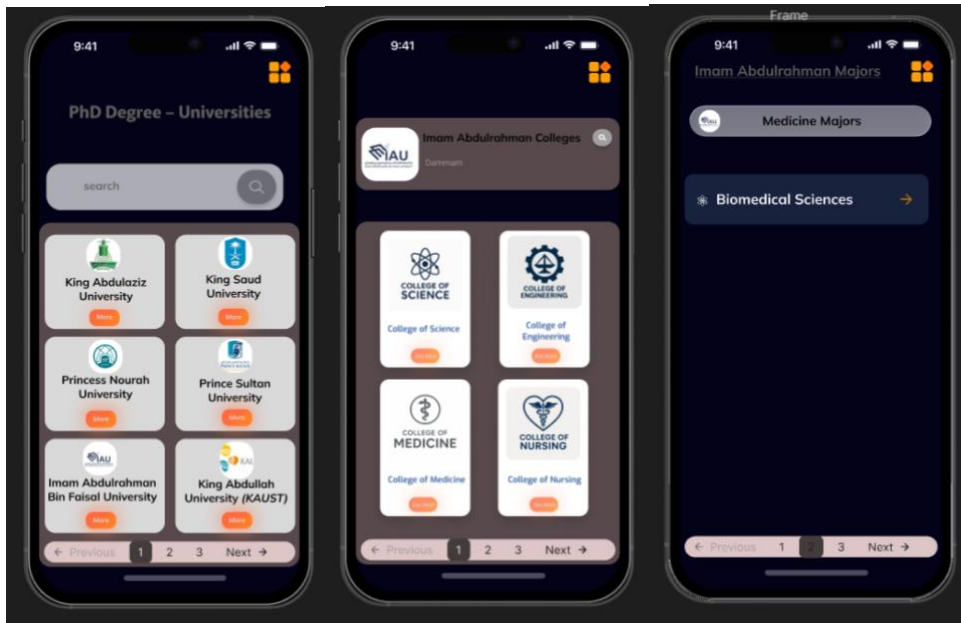


Figure 7 User interface screens showing the process of browsing PhD universities, exploring available colleges, and viewing majors under the selected college.

5. If the user clicks on setting icon:

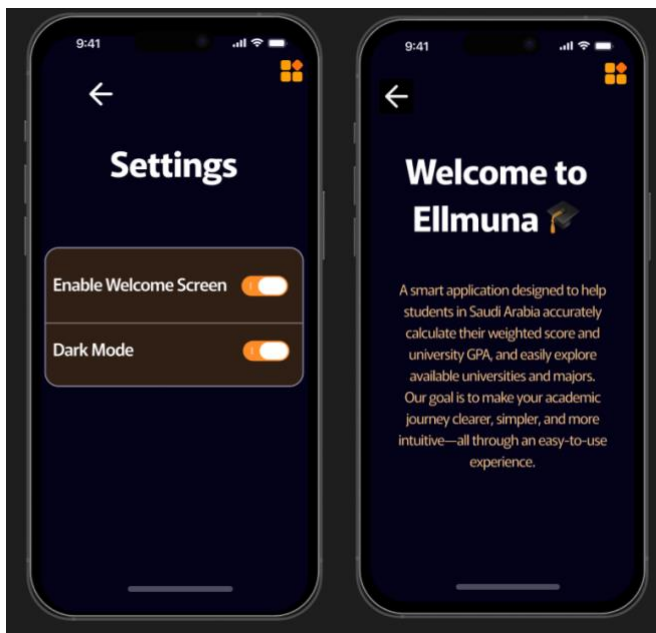


Figure 8 User interface screens for accessing the Settings page and viewing the Welcome Screen.

4.4 Screen Objects and Actions



Interface Description

The **Home Screen** serves as the main entry point of the Ellmuna application. The top section displays the application logo, which represents the visual identity of the system. Below the logo, three main buttons provide quick access to the core features:

1. **Weighted Average Calculator** for high school students.
2. **GPA Calculator** for university students.
3. **Saudi Universities** browsing section.

A **menu icon** is placed at the top-right corner, allowing users to open a side navigation panel with additional settings and options.

Screen Objects

Object	Description
Application Logo	Represents the app branding, displayed at the top of the screen.
Weighted Average Calculator Button	Navigates to the weighted score calculator screen.
GPA Calculator Button	Opens the university GPA calculation interface.
Saudi Universities Button	Leads to the section for browsing Saudi

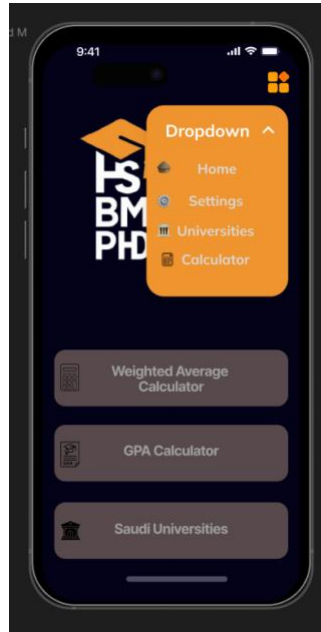
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	universities and programs.
Menu Icon	Opens the side navigation menu for additional settings.

Actions

Object	Action
Weighted Average Calculator Button	Redirects the user to the weighted score input screen.
GPA Calculator Button	Opens the page for choosing GPA system and entering previous GPA data.
Saudi Universities Button	Displays the university exploration module (Bachelor, Master, PhD).
Menu Icon	Opens the side navigation drawer.

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Interface Description

This interface displays the dropdown menu that appears when the user taps the menu icon in the top-right corner. The dropdown provides quick navigation to the main sections of the application without returning to the Home Screen. At the top, the menu title *Dropdown* is shown, along with a toggle arrow to collapse the menu. Below it, several navigation options are listed, each with an identifying icon for clarity and better user experience.

Screen Objects

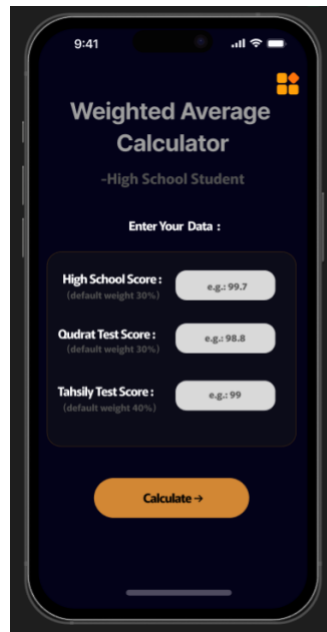
Object	Description
Menu Title (Dropdown)	Indicates that the dropdown is open and allows collapsing when the arrow is tapped.
Home Button	Navigates to the main Home Screen.
Settings Button	Opens the Settings page for user preferences like night mode and welcome screen toggle.
Universities Button	Opens the list of Saudi universities by degree level.
Calculator Button	Redirects to the calculation tools (Weighted

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	Average or GPA).
--	------------------

Actions

Object	Action
Home	Redirects to the Home Screen.
Settings	Opens the Settings interface.
Universities	Opens the browsing interface for Saudi universities.
Calculator	Redirects to the calculator selection page.
Arrow Icon	Closes and collapses the dropdown menu.



Interface Description

This screen provides the input form required to calculate the weighted average score for high school students. At the top, a clear title identifies the purpose of the screen, followed by a subtitle “High School Student.” A central input card contains three primary fields: High School Score, Qudrat Test Score, and Tahsily Test Score, each labeled with its default weight. Each field includes a text input box on the right side with a placeholder example to guide the user. At the bottom, the “**Calculate**” button processes the entered data and navigates the user to the result screen.

Screen Objects

Object	Description
Screen Title	Displays the purpose of the interface (Weighted Average Calculator)
Subtitle (High School Student)	Identifies the user category
Input Card	Contains the three main score fields
High School Score Field	For entering the student’s school percentage (with default weight 30%)

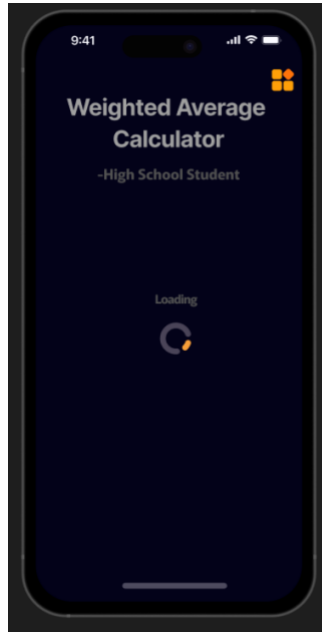
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Qudrat Test Score Field	For entering the aptitude test score (default weight 30%)
Tahsily Test Score Field	For entering the achievement test score (default weight 40%)
Placeholders (e.g., 99.7)	Provide examples to guide proper input
Calculate Button	Executes the weighted average calculation

Actions

Object	Action
Score Input Fields	User enters numeric values; system validates input
Calculate Button	Performs weighted score calculation and navigates to result screen
Placeholders	Disappear when the user starts typing

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Interface Description

The *Loading Screen* serves as a transitional interface within the Weighted Average Calculator module. It appears immediately after the user enters their required data and taps on the “Calculate” button. The purpose of this screen is to inform the user that the system is currently processing the input and calculating the final weighted average, ensuring a smooth experience and preventing duplicate interactions. The screen displays a centered loading spinner along with the text “Loading” to visually indicate that the calculation is in progress. No interactive elements are available on this screen, and the system automatically redirects the user to the results page once the computation is complete.

Screen Objects

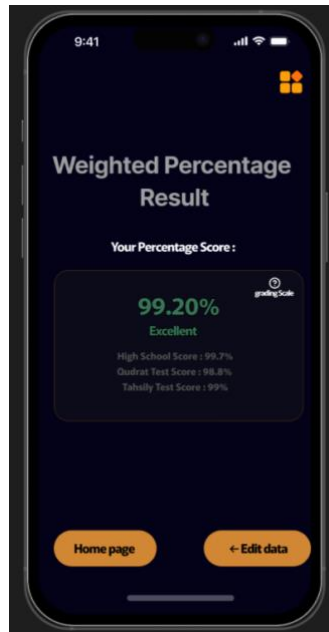
Object	Description
Screen Title	Shows “Weighted Average Calculator – High School Student” to indicate context
Loading Text	Displays “Loading” to notify the user that processing is ongoing
Loading Spinner	Visual indicator showing that the system is working in the background
Menu Icon	Visible at the top, but not intended for interaction during the loading state

Actions

Object	Action
System (Backend Process)	Processes the user’s data and calculates the weighted score
Loading Screen	Locks user input to avoid interruptions and

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	redirects automatically to the results page
Menu Icon	Remains visible but does not affect the calculation process



Interface Description

This screen displays the final weighted percentage result for the high school student. At the top, the header “Weighted Percentage Result” is shown, followed by a subtitle indicating that the displayed value is the student’s final score.

The main result is presented inside a central card that includes:

- The final percentage (large and highlighted).
- The academic evaluation (e.g., Excellent).
- A detailed breakdown of the student’s three scores: High School, Qudrat, and Tahsily.
- A small icon at the top corner to open the **grading scale** window.

At the bottom, two primary action buttons are provided:

1. **Edit data** to return and modify the entered scores.
2. **Home page** to navigate back to the main screen.

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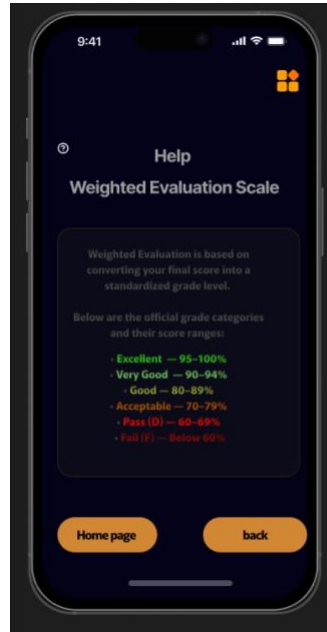
Screen Objects

Object	Description
Screen Title	Main header: <i>Weighted Percentage Result</i>
Subtitle	Indicates that the following value is the student's percentage score
Result Card	Contains percentage, evaluation, and detailed scores
Final Percentage (99.20%)	Highlighted final weighted score
Evaluation Text	Displays the academic level (e.g., Excellent)
Grades Breakdown	Shows High School, Qudrat, and Tahsily values
Grading Scale Icon	Opens the grading scale for reference
Edit Data Button	Returns to previous screen to modify inputs
Home Page Button	Navigates back to main menu

Actions

Object	Action
Grading Scale Icon	Opens grading scale for review
Edit Data	Returns to the data input screen
Home Page	Redirects to the Home Screen

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Interface Description

This screen provides students with a clear explanation of the weighted evaluation scale, helping them understand how their final weighted score is categorized. At the top, the screen displays a help icon along with the main and secondary titles. The center of the screen contains a styled information box that explains how weighted evaluation works and lists the official grade categories (Excellent, Very Good, Good, etc.) with their corresponding score ranges. This interface aims to help users interpret their results easily and accurately.

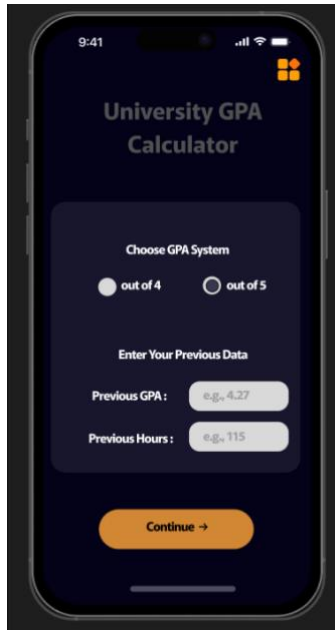
Screen Objects

Object	Description
Help icon (?)	Indicates that this is a help/guide section
Main title "Help"	Identifies the page as a help screen
Subtitle "Weighted Evaluation Scale"	Specifies that the page explains the grading scale
Information Box	Contains explanation text and the colored grading scale list
Home page Button	Navigates back to the main screen
Back Button	Returns the user to the previous result screen

Actions

Object	Action
Home page	Redirects to the Home Screen
Back	Goes back to the previous screen
Help icon	Display-only (no action)

Software Design Specifications for Ellmuna App



Interface Description

This screen presents the university GPA calculator, allowing the user to select their university's GPA system (out of 4 or out of 5). The user can then enter their previous academic information, including their previous GPA and previous credit hours. The interface is simple, clean, and provides guiding placeholders to ensure correct data entry. After filling in the required fields, the user can proceed to the next step to add current semester courses.

Screen Objects

Object	Description
Page Title – University GPA Calculator	Identifies the purpose of the screen.
GPA System Selection Box	Allows the user to choose between “out of 4” or “out of 5.”
Radio Buttons	Interactive elements used to select only one GPA system.
Previous GPA Field	Input field for entering the previous GPA with an example placeholder.
Previous Hours Field	Input field for the user's previously completed credit hours.

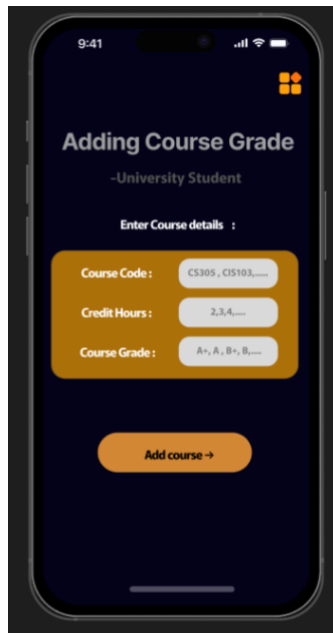
Software Design Specifications for Ellmuna App

Continue Button	Navigates the user to the next screen for detailed GPA calculations.
-----------------	--

Actions

Object	Action
Out of 4	Sets GPA system to a 4-point scale.
Out of 5	Sets GPA system to a 5-point scale.
Previous GPA Input	Allows numeric entry of the user's previous GPA.
Previous Hours Input	Allows numeric entry of previous credit hours.
Continue Button	Moves to the course-entry screen for GPA computation.

Software Design Specifications for Ellmuna App



Interface Description

This screen allows university students to enter the details of a single course as part of the GPA calculation process. The user inputs the course code, credit hours, and the grade achieved. After entering the required information, the student presses the **Add Course** button to add the course to the temporary list of courses before calculating the semester GPA or cumulative GPA. The interface is designed to be simple, clean, and user-friendly to support quick data entry.

Screen Objects

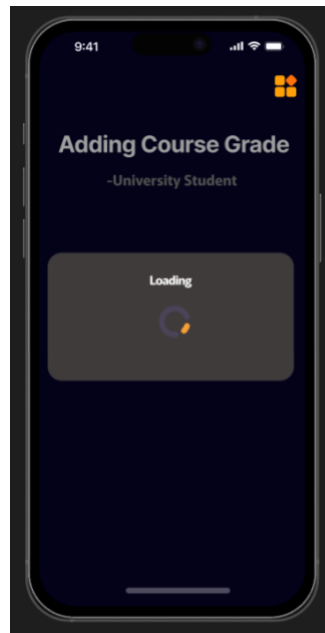
Object	Description
Course Code Input	Input field for entering course code (e.g., CS305, CIS103...)
Credit Hours Input	Field for entering the credit hours (2, 3, 4...)
Course Grade Input	Field/dropdown for selecting the grade (A+, A, B+...)
Add Course Button	Button used to add the course to the user's course list
Menu Icon	Top-right icon used to open the navigation dropdown

Actions

Software Design Specifications for Ellmuna App

Object	Action
Course Code Input	Allows the user to type the course code
Credit Hours Input	Allows entering credit hours
Course Grade Input	Allows selecting or typing the course grade
Add Course Button	Adds the entered course to the list and saves inputs temporarily
Menu Icon	Opens the navigation menu

Software Design Specifications for Ellmuna App



Interface Description

This screen represents the loading phase that appears after the user taps the “**Add course**” button in the course grade submission interface. It indicates that the system is processing the provided data (course code, credit hours, and course grade). The screen displays a rotating loading indicator and a simple message to keep the user informed about the ongoing background operation.

Screen Objects

Object	Description
Title Area	Displays the title “Adding Course Grade – University Student” for consistency with the previous screen
Loading Message	Shows the text “Loading” to indicate that data processing is in progress
Loading Indicator	A spinning circular animation that visually communicates system activity

Actions

Object / Process	Action
Loading Indicator	Non-interactive; visually informs the user
System Processing	The backend processes the input data (course code, credits, grade)
Auto Navigation	System automatically redirects to the next screen once processing is complete

Software Design Specifications for Ellmuna App



Interface Description

The **Courses Details** screen displays all university courses added by the student before calculating the GPA. This interface organizes the previously entered courses and allows the student to verify all details, including:

- **Course Code**
- **Credit Hours**
- **Course Grade**

The screen includes three main action buttons:

- **Calculate GPA:** Computes the final GPA.
- **Add Course:** Allows adding additional courses.
- **Home Page:** Returns to the main dashboard.

Screen Objects

Object	Description
Courses Details Title	Indicates that the user is viewing the list of added courses
Semester Courses Section	Displays the list of the semester's courses
Course Code Column	Shows each course's official code
Credit Hours Column	Displays credit hours for each course
Course Grade Column	Shows the earned grade for each course
Calculate GPA Button	Starts the GPA calculation process

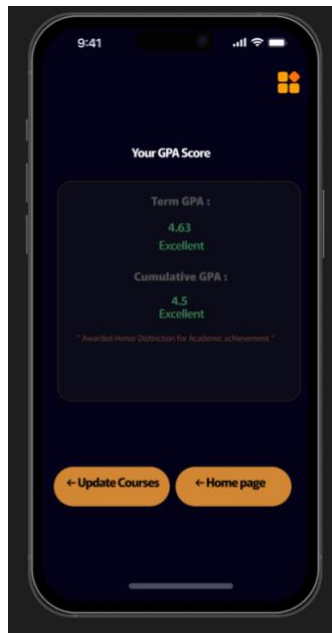
Software Design Specifications for Ellmuna App

Add Course Button	Navigates to the “Add Course” screen
Home Page Button	Returns user to the home interface

Actions

Object	Action
Calculate GPA	Proceeds to the GPA result screen
Add Course	Opens the course-adding page
Home Page	Navigates back to the Home screen

Software Design Specifications for Ellmuna App



Interface Description

This screen displays the final calculated results for both the **Term GPA** and the **Cumulative GPA** for university students. After entering their course data, the user is shown their GPA results along with an academic classification (e.g., Excellent, Very Good). The interface includes a main results box, a highlight note for academic distinction if applicable, and two navigation buttons: one to return to the home page and another to update or modify course entries.

Screen Objects

Object	Description
Page Title (Your GPA Score)	Indicates that the screen displays GPA results.
Term GPA Box	Shows the term GPA along with its academic classification.
Cumulative GPA Box	Displays the cumulative GPA with its corresponding classification.
Achievement Note	Appears when the GPA qualifies for academic honors.
Update Courses Button	Allows the user to return and modify course data.
Home Page Button	Takes the user back to the main home screen.

Actions

Object	Action
Term GPA Box	Displays the user's term GPA and

Software Design Specifications for Ellmuna App

	classification.
Cumulative GPA Box	Displays the final cumulative GPA.
Update Courses Button	Navigates to the course editing interface.
Home Page Button	Navigates back to the Home screen.

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Interface Description

This screen is the entry point for exploring Saudi universities within the *Ellmuna* application. It allows the user to select the academic degree level they wish to browse (Bachelor's, Master's, or PhD). The interface is designed to be simple and intuitive, presenting three clear action buttons representing each degree level.

A small header bar with an icon is displayed at the top to visually indicate that the user is selecting a degree level.

Screen Objects

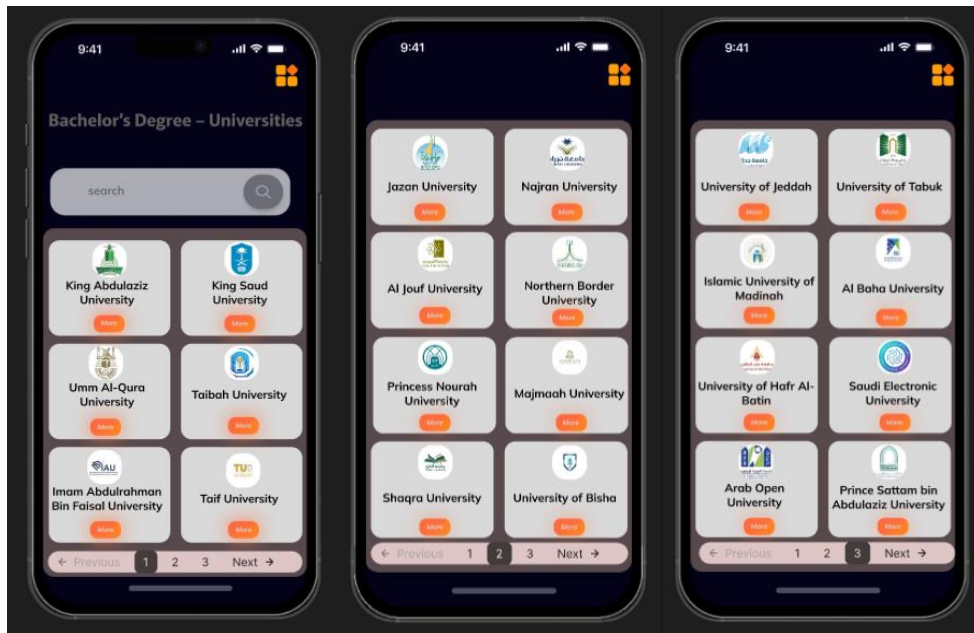
Object	Description
Degree Level Header Bar	Informs the user that they are choosing the academic level for browsing.
Bachelor's Degree Button	Navigates to the list of universities offering Bachelor's programs.
Master's Degree Button	Opens the list of universities offering Master's programs.
PhD Degree Button	Opens the list of universities offering PhD programs.
Menu Icon	Opens the side navigation menu with additional options.

Actions

Software Design Specifications for Ellmuna App

Object	Action
Bachelor's Degree	Navigates to Bachelor-level university list.
Master's Degree	Opens Master's university list.
PhD Degree	Opens PhD-level university list.
Menu Icon	Opens navigation drawer for more options.

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Interface Description

This interface displays all Saudi universities that offer Bachelor's degree programs. It provides a clean and organized browsing experience through a search bar located at the top, allowing users to quickly find a specific university. Below the search area, universities are presented in card-style boxes that include the university logo and its name. Each card contains a *More* button that directs the user to a detailed page showing the available colleges and majors within that university. At the bottom, pagination controls allow the user to navigate between multiple pages of universities.

Screen Objects

Object	Description
Search Bar	Allows the user to enter a university name.
Search Icon	Executes the search operation.
University Card	Displays the university's logo and name.
More Button	Opens a detailed page showing colleges and majors.
Pagination Bar	Enables navigation between list pages (Previous – Next – Page Number).

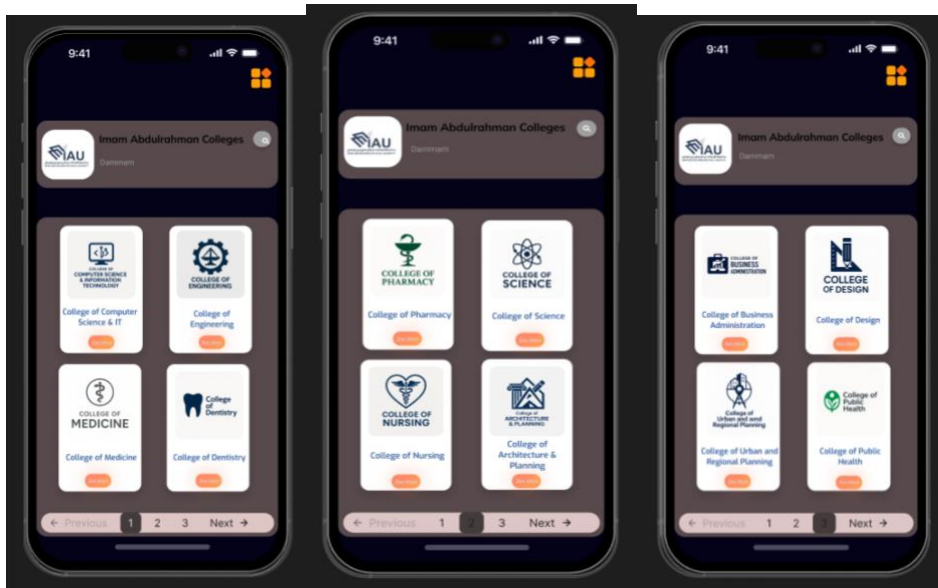
Actions

Object	Action
Search Bar	User enters a name to filter universities.
Search Icon	Executes the search and displays results.

Software Design Specifications for Ellmuna App

University Card	Opens the selected university information.
More Button	Leads to university colleges & majors page.
Pagination Controls	Moves to the next or previous set of universities.

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Description

This screen displays the list of colleges within **Imam Abdulrahman Bin Faisal University (IAU)**. The university information appears at the top in a card that includes the university logo, name, and location. Below that, a grid of college cards is displayed—each showing the college logo and its title to allow users to browse easily. A search field is provided at the top of the colleges section to help users find a specific college quickly. Pagination controls at the bottom allow users to navigate between multiple pages of college listings.

Screen Objects

Object	Description
University Info Card	Displays the university logo, name, and location.
Search Bar	Allows the user to search for a specific college.
College Cards	Each card shows a college logo, its name, and a “See More” button.
Pagination Controls	Buttons to move between pages (Next / Previous / Page numbers).

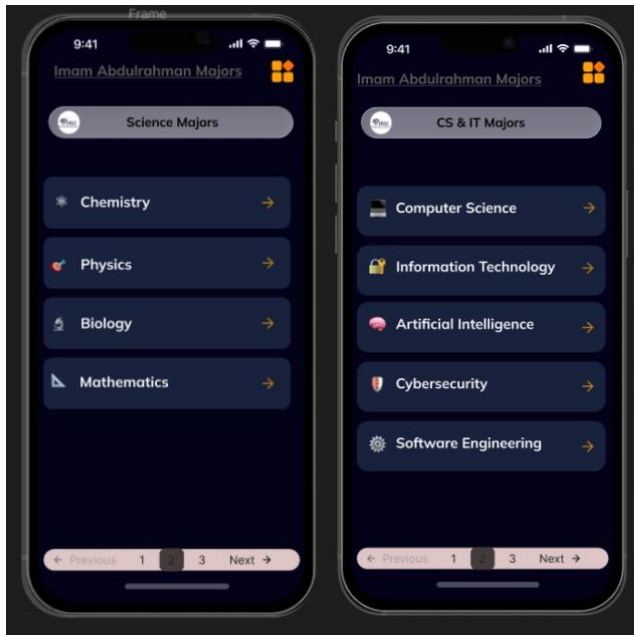
Actions

Object	Action
College Card	Opens the detailed page showing majors within that college.
Search Bar	Filters colleges based on user input.

Software Design Specifications for Ellmuna App

Next / Previous	Navigates through college pages.
See More Button	Opens college details page.

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Interface Description

The *Science Majors* screen displays all scientific majors available under the selected university's College of Science. This interface helps students browse majors easily and navigate directly to the major they are interested in. At the top, the screen shows the university name and the college label, providing context for the user. Below, a list of major cards appears, each representing a scientific major such as Chemistry, Physics, Biology, or Mathematics. Every card includes an icon and an arrow indicating navigation to the detailed major page. Pagination controls appear at the bottom of the screen, enabling users to move between pages when the list is long.

Screen Objects

Object	Description
University + College Title	Displays the university and college currently being viewed.
“Science Majors” Header	Indicates the category being displayed.
Major Cards	Each card represents one scientific major.
Arrow Icon	Opens the major’s detailed page.
Pagination Bar	Allows movement across multiple pages of majors.

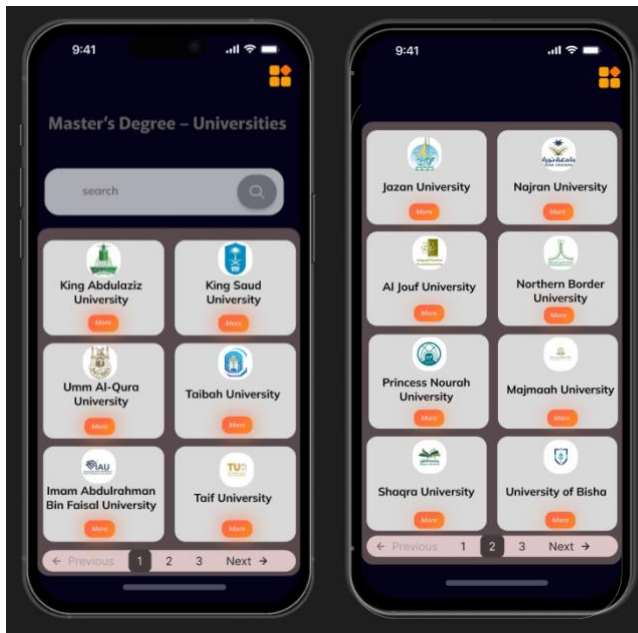
Actions

Object	Action
Chemistry Button	Navigates to Chemistry major details.
Physics Button	Navigates to Physics major details.

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Biology Button	Opens the Biology major page.
Mathematics Button	Opens the Mathematics major page.
Pagination (Previous – Next – Numbers)	Moves to the next or previous list of majors.

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Interface Description

This screen displays a list of Saudi universities that offer Master's degree programs. Users can search for a specific university using the search bar at the top, then browse through the displayed university cards, each containing the university logo, its name, and a “More” button to view detailed programs. A pagination bar is provided at the bottom to switch between pages. A side menu icon is also available at the top-right corner for additional navigation.

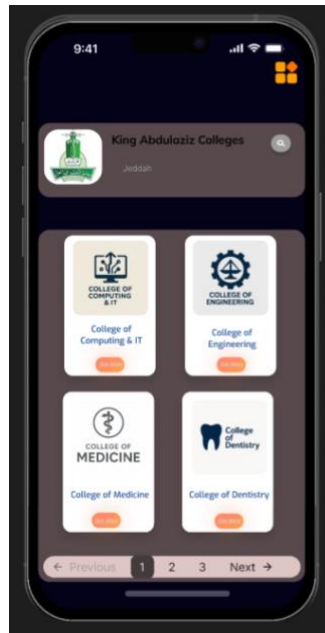
Screen Objects

Object	Description
Search Bar	Allows users to search for a specific university by name
University Card	Displays the university logo and name
“More” Button	Navigates to the university's colleges and Master's programs
Pagination Bar	Used to switch between pages of universities
Menu Icon	Opens the navigation drawer

Actions

Object	Action
Search Bar	Filters universities based on keywords entered
“More” Button	Redirects to detailed university programs
Pagination	Moves to the next/previous page of results
Menu Icon	Opens the side navigation menu

Software Design Specifications for Ellmuna App



Description

At the top, a university card displays:

- King Abdulaziz University logo
- University name and location (Jeddah)
- A search icon that allows the user to look for a specific college.

Below the header card, multiple square cards represent the university colleges. Each card includes:

- The college logo
- The college name
- A small “See More” button to show details of the programs offered.

At the bottom, pagination is available to move between pages 1, 2, and 3.

Screen Objects

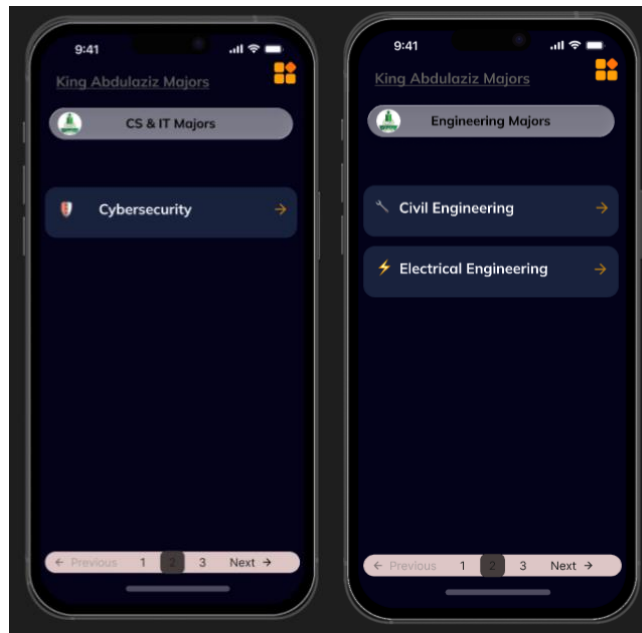
Object	Description
University header card	Shows university logo, name, location, and search icon
Search Icon	Allows user to search for a specific college
College cards	Each represents a college with its logo and name
“See More” button	Opens the list of programs available in that college
Pagination	Allows navigation between pages of colleges

Software Design Specifications for Ellmuna App

Screen Actions

Action Object	Action
Search icon	Opens input field for college search
College card	Navigates to the list of majors offered in that college
See More button	Shows detailed information about the college programs
Pagination buttons	Switch pages to view more colleges

Software Design Specifications for Ellmuna App



Interface Description

This screen displays the available majors within **King Abdulaziz University**, specifically under the selected college (*CS & IT Majors*). The header shows the university name, followed by a college selection bar. Below that, the user sees the list of majors offered within that college. In this interface, only one major—**Cybersecurity**—is displayed, each with an arrow button to navigate to detailed information.

Screen Objects

Object	Description
University Name (King Abdulaziz Majors)	Indicates which university is currently displayed.
College Bar – CS & IT Majors	Shows the selected college within the university.
Major Card (Cybersecurity)	Displays the available major with a related icon.
Right Arrow (→)	Navigates to the detailed information page of the major.
Pagination Bar (Previous – 1 – 2 – 3 – Next)	Allows navigation between major pages.

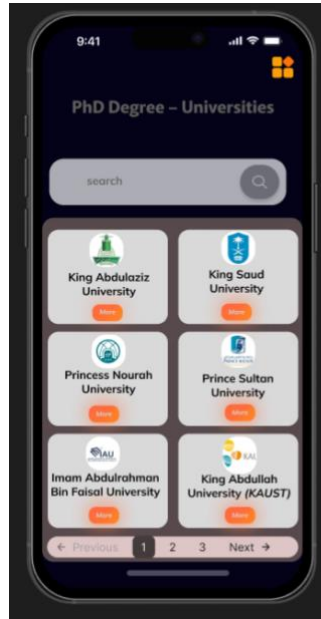
Actions

Object	Action
Cybersecurity Card	Opens the major details page.
Previous / Next	Moves between major pages.

Software Design Specifications for Ellmuna App

Page Numbers	Directly jumps to a selected page.
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Description

This interface displays all Saudi universities that offer **PhD programs**. At the top, a search bar allows users to quickly locate a specific university. Below the search bar, universities are presented as individual cards containing the university logo, name, and a “More” button for additional details. The interface also includes a pagination bar at the bottom (Previous – page numbers – Next) for browsing through multiple pages of universities.

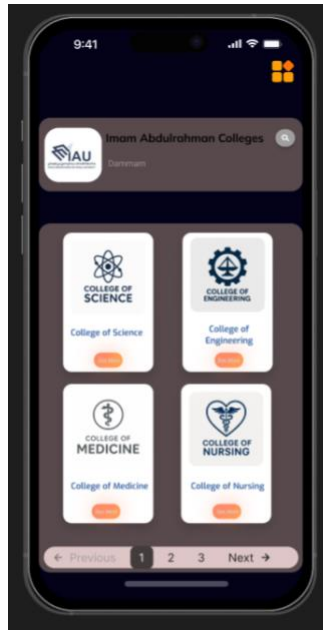
Screen Objects

Object	Description
Search Bar	Allows users to type and search for a specific university.
University Cards	Display the university’s logo, name, and a “More” button.
Pagination Bar	Enables navigation between multiple pages of universities.
Menu Icon	Opens the top dropdown navigation menu.

Actions

Object	Action
Search Bar	Filters the displayed universities based on user input.
More Button	Navigates to the selected university’s college list.
Pagination Buttons	Moves between university pages.
Menu Icon	Opens the additional navigation menu.

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Interface Description

This interface displays all colleges under **Imam Abdulrahman Bin Faisal University (IAU)** after the user selects the university from the main list. At the top, a header card shows the **university logo, name, and city**, along with a **search icon** for filtering colleges.

Below the header, a grid of college cards is shown. Each card contains:

- College logo
- College name
- A “See More” button to navigate to the list of majors offered in that college

At the bottom, a **pagination bar** allows the user to navigate across pages if the number of colleges is large.

Screen Objects

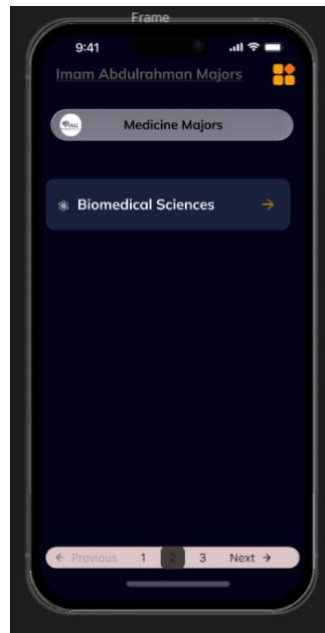
Object	Description
University Header Card	Displays the university logo, name, and city
Search Icon	Opens search field to filter colleges
College Cards	Cards showing the logo and name of each college
“See More” Button	Opens the list of majors inside the selected college
Pagination Bar	Allows navigation between pages

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Actions

Object	Action
Search Icon	Opens a search functionality
College Card	Navigates to the college's majors page
See More Button	Opens detailed information and majors list
Pagination Buttons	Moves between pages (Previous – 1 – 2 – 3 – Next)

Software Design Specifications for Ellmuna App



Description

The **Medicine Majors** screen displays the academic majors available under the College of Medicine at **Imam Abdulrahman Bin Faisal University**. At the top, a breadcrumb navigation link allows users to return to the previous page (university colleges list). Below it, a header bar shows the college name “*Medicine Majors*” along with the university logo for clear academic identification. The main section presents the available major—**Biomedical Sciences**—inside a highlighted card that includes a relevant medical-themed icon and a navigation arrow leading to the major details page. This interface helps users browse medical programs easily and understand the available academic paths within the college.

Screen Objects

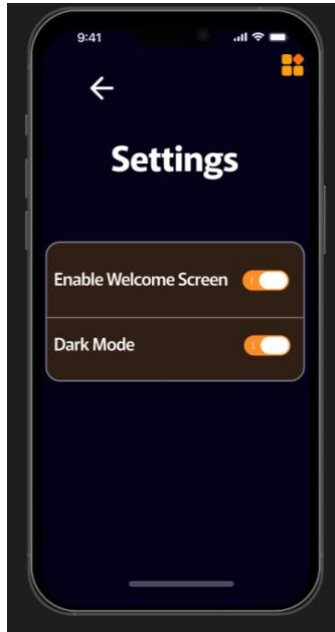
Object	Description
Breadcrumb Navigation	Allows the user to return to the previous page in the hierarchy.
College Header	Displays the university logo and the college name “Medicine Majors”.
Major Card	Shows the available major: Biomedical Sciences .
Navigation Arrow	Leads to the detailed page of the selected major.
Pagination Bar	Enables navigating between multiple pages (Previous – 1 – 2 – 3 – Next).

Actions

Software Design Specifications for Ellmuna App

Object	Action
College Header	Visual identification of the college category.
Biomedical Sciences Card	Redirects to the Biomedical Sciences major details page.
Breadcrumb	Takes the user back to the College List page.
Pagination	Moves between pages of majors or colleges.

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Interface Description

The Settings screen allows users to customize basic application preferences within *Ellmuna*. The interface presents two main toggle options that let users enable or disable certain features based on their needs. A back arrow at the top-left corner helps users return to the previous screen easily.

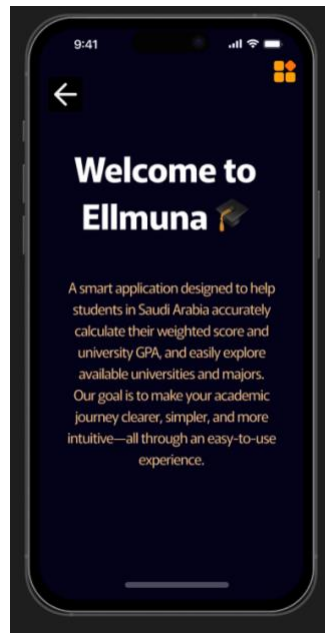
Screen Objects

Object	Description
Back Arrow (←)	Navigates back to the previous page.
Page Title “Settings”	Indicates that the user is inside the Settings section.
Enable Welcome Screen Option	Allows turning the welcome screen on or off.
Dark Mode Option	Allows switching between dark and light themes.
Toggle Switch	Interactive element for enabling/disabling each setting.

Actions

Object	Action
Back Arrow ←	Returns to the previous interface.
Enable Welcome Screen	ON: the welcome screen appears when opening the app. OFF: app bypasses the welcome screen.
Dark Mode	ON switches the UI to dark theme; OFF returns to light theme.
Toggle Switch	Immediately changes the setting and saves it automatically.

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Interface Description:

The Welcome Screen provides a friendly introduction to the Ellmuna application. It immediately informs the user about the purpose of the app through a clear title and a descriptive paragraph. At the top, a back arrow allows the user to return to the previous page, while the menu icon on the top-right provides access to additional navigation options. At the center, a bold welcome title “Welcome to Ellmuna” along with a graduation cap icon introduces the app’s identity. Below it, a descriptive paragraph explains that the application helps students in Saudi Arabia calculate their weighted average, university GPA, and explore available universities and majors easily and intuitively. This screen creates a smooth entry point and sets clear expectations for the user before they start interacting with the system.

Screen Objects

Object	Description
Back Button	Returns user to the previous screen
Welcome Title	Introduces the app and greets the user
Description Text	Explains the main purpose and functions of the app
Menu Icon	Opens side navigation menu

Actions

Object	Action
Back Button	Navigates to the previous screen
Menu Icon	Opens the navigation drawer

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Description Text	Static element; no action
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4.5 Other Interfaces

The *Ellmuna* application includes additional interfaces that support internal user interaction. These interfaces use different UI mechanisms to display data or enhance user experience.

1. Welcome Screen Interface

- **Technology:** Front-end interactive UI layout.
- **Interaction:** Appears only if “Enable Welcome Screen” is activated.
- **Message Type:** Informational text describing the purpose of the application.
- **Closure:** Closed manually through the back button.

2. Side Navigation Menu

- **Technology:** Dropdown navigation interface.
- **Interaction:** Opens/closes through the menu icon.
- **Protocol:** Internal navigation between screens.
- **Error Handling:** Not required as no data input is involved.

3. Loading Interfaces

- **Purpose:** Indicate data processing before showing results.
- **Technology:** Animated loader.
- **Closure:** Automatically dismissed after processing is completed.

4. Result Feedback Interfaces

- **Interaction:** Displays the user’s weighted score or GPA with performance evaluation.
- **Data Format:** Numeric results + textual classification.
- **Error Conditions:** Triggered only when required inputs are missing or invalid (validated internally).

5. System Architecture

In this section, a high-level overview of the system architecture for the Ellmuna Mobile Application is presented. The goal is to explain how the system's functionalities and responsibilities were partitioned into subsystems, how these subsystems interact, and the rationale behind choosing the adopted architectural style. The description focuses on overall structure rather than low-level details, which will be covered in the Detailed Design section.

5.1 Architectural Design Approach

The optimal architectural design approach selected for the Ellmuna mobile application is the Layered Architecture (Three-Tiered Model). This approach is the most suitable for mobile applications that require clear separation between user interaction, system logic, and stored preferences. Each functionality in the system is assigned to its appropriate layer, ensuring that any future updates or modifications will affect only the targeted layer without impacting the entire system, which increases maintainability and reduces system complexity.

In addition, the layered approach minimizes module dependency, allowing the team to develop UI components, calculation logic, and data handling independently. This is ideal for a mobile system intended to run consistently across different devices and operating systems. Moreover, choosing a layered architecture enhances the system's reliability and improves the clarity of communication between components, which is essential for handling user preferences, GPA calculations, and browsing university programs.

Furthermore, compared to a non-layered approach, layered architecture provides better structure and scalability. A non-layered model would make the system harder to maintain, more vulnerable to errors, and more difficult to extend with new features such as additional academic programs or new calculation tools.

The Ellmuna system architecture consists of three main layers:

1. Presentation Layer (UI Layer)
2. Business Layer (Application Logic Layer)
3. Data Layer (Local Storage Layer)

In comparison to layered approach, the implementation of a non-layered approach will make the system exposed to external threats and more likely to suffer from protection issues. Moreover,

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implementing portable and changeable software will be extremely challenging comparatively.

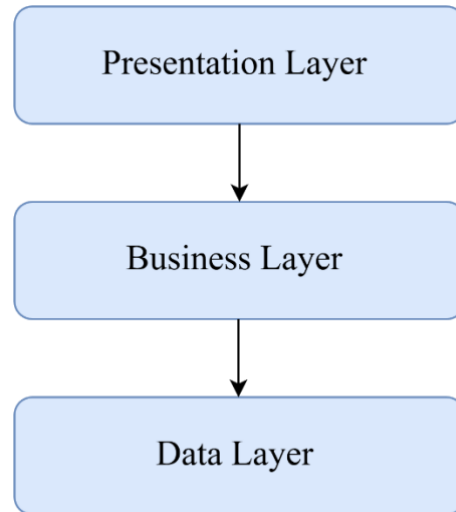


Figure 9 Ellmuna App Layered Approach

5.2 Architectural Design

The overall system architecture of the Ellmuna mobile application is illustrated in the following diagram, which represents the high-level structure of the system and the main subsystems that compose it. The architecture is designed in a modular way to ensure clarity, maintainability, and scalability as additional features or academic programs may be added in the future.

The modular program structure divides the system into several high-level subsystems, each responsible for a specific set of functionalities. These subsystems interact with each other to achieve the complete system behavior and ensure smooth data flow throughout the application.

High-Level Subsystems

Presentation Layer (UI Subsystem)

This subsystem handles all user-facing components, including the screens for home navigation, GPA calculation, weighted percentage calculation, browsing universities, and managing settings. It collects user input and displays processed results. It acts as the entry point to the system and forwards user interactions to the appropriate subsystem.

Application Logic Layer (Processing Subsystem)

This subsystem performs the core processing tasks of the application. It includes the logic for weighted percentage calculations, GPA calculations (semester & cumulative), validating user inputs, and filtering universities and academic programs. It serves as the central coordinator between the UI layer and the Data Management subsystem.

Data Management Layer (Data Repository Subsystem)

This subsystem handles all local data storage and retrieval. It is responsible for storing user preferences (such as dark mode and welcome screen settings) and maintaining static datasets for universities, colleges, and majors. It provides required data to the Application Logic subsystem upon request.

Subsystem Collaboration

These subsystems work together to provide the complete functionality of the application:

- The Presentation Layer captures user actions and sends the required data to the Application Logic Layer.
- The Application Logic Layer processes these actions, performs calculations, applies filters, and requests stored data from the Data Management Layer when needed.
- The Data Management Layer retrieves and returns data back to the Application Logic Layer.
- Finally, the Application Logic Layer returns processed results to the Presentation Layer for display to the user.

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This collaboration ensures a well-organized structure with clear boundaries between responsibilities, improving the system's scalability, maintainability, and reliability.

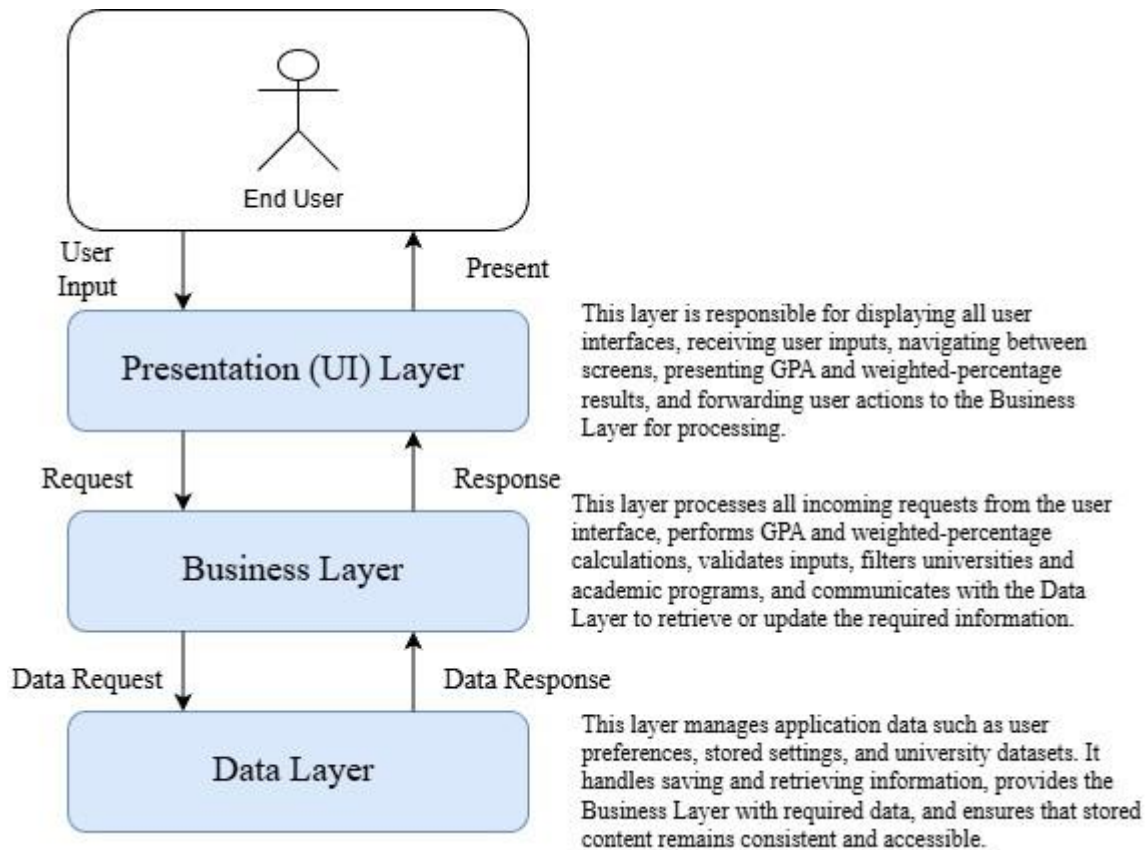


Figure 10 Ellmuna App Architecyure Design

5.3 Subsystem Architecture

The Ellmuna mobile application is decomposed into several subsystems to improve modularity, simplify maintenance, and ensure clear separation of responsibilities. Each subsystem handles a specific group of related functionalities and interacts with other subsystems through well-defined data flows.

This structure supports scalability and allows different system components—such as calculations, university browsing, and user preferences—to be developed and extended independently without affecting the overall system.

5.3.1 General View of the System

The context diagram below provides a high-level Data Flow Diagram (DFD) of the Ellmuna Application. It shows how different external entities interact with the system, what type of data they send to the system, and what information the system returns back to them.

- High School Student → sends score inputs to the system and receives weighted-percentage results.
- University Student → submits GPA information and retrieves calculated GPA results.
- Universities & Majors Database → provides academic data (majors, universities) and receives requests from the system.

The diagram illustrates the system boundaries clearly and presents how data flows between external users and the Ellmuna App.

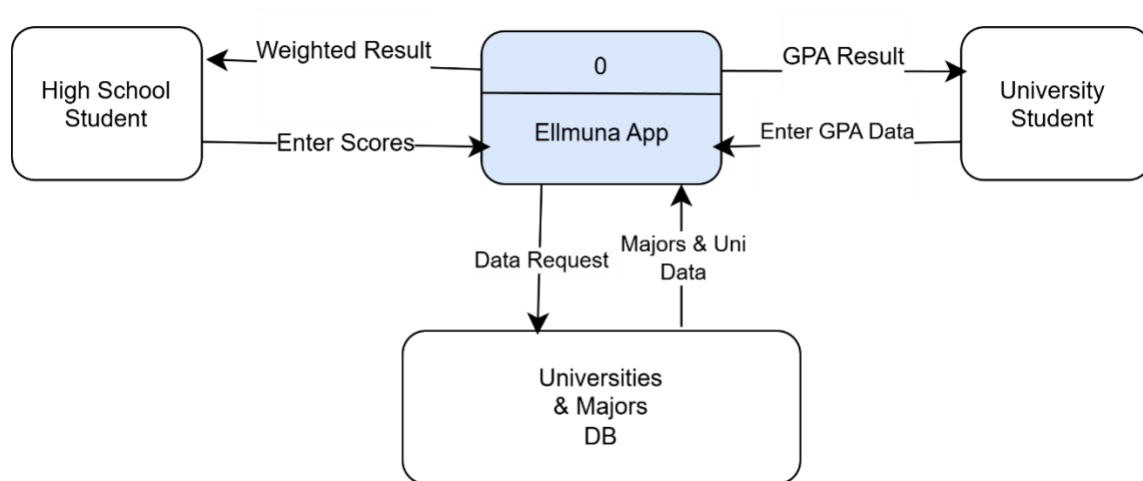


Figure 11 Context DFD of Ellmuna App

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5.3.2 User's Subsystem

In this subsection, a Level-0 Data Flow Diagram (DFD) is provided to illustrate the common functionalities offered to both types of users in the Ellmuna Application. These functionalities focus on academic-related calculations that support students in evaluating their academic performance.

Two main processes are shared across the system:

- **1.0 Calculate Weighted Percentage:**
This process is used by *High School Students* to submit their exam scores and receive the calculated weighted percentage result.
- **2.0 Calculate GPA:**
This process is used by *University Students* to enter their GPA-related information and retrieve their computed GPA result.

The Level-0 DFD shown below demonstrates how data flows between external users and the Ellmuna App while performing these two essential functionalities. It clarifies what information users send to the system and what results the system returns back.

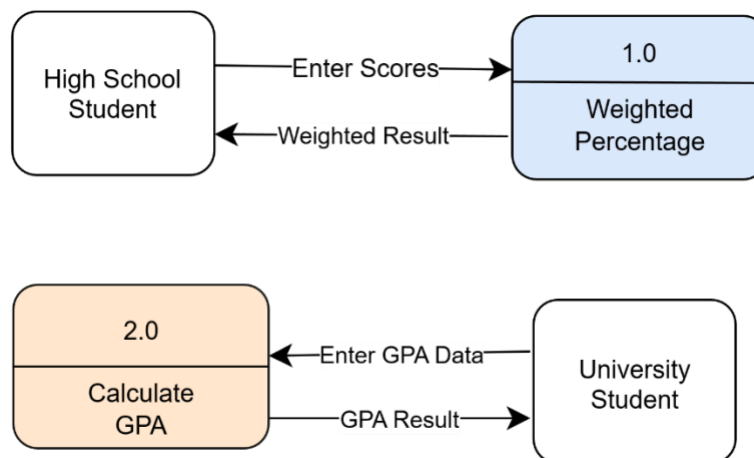


Figure 12 Level 0 – DFD of Common Subsystem Functionalities

5.3.3 Universities & Majors Subsystem

This subsystem is responsible for providing users with academic information about Saudi universities and the majors offered at different academic levels. It supports both high-school and university students by allowing them to explore educational opportunities and understand available specializations.

Two primary processes are included in this subsystem:

• 3.0 Browse Universities

The user selects an academic level (Bachelor, Master, or PhD).

The system sends a request to the *Universities & Majors Database* to retrieve all universities that offer programs at the selected level.

The list of universities is then displayed to the user.

• 4.0 Browse Majors

Upon selecting a specific university (and a college for Bachelor level), the user initiates a request to view the majors available within that selection.

The system queries the database and returns a list of majors relevant to the chosen university and academic level.

This subsystem plays a key role in the Ellmuna App, as it provides essential academic information that helps users explore educational paths and identify suitable programs. The Level-0 DFD below illustrates the data flow between the user, the subsystem processes, and the academic database.

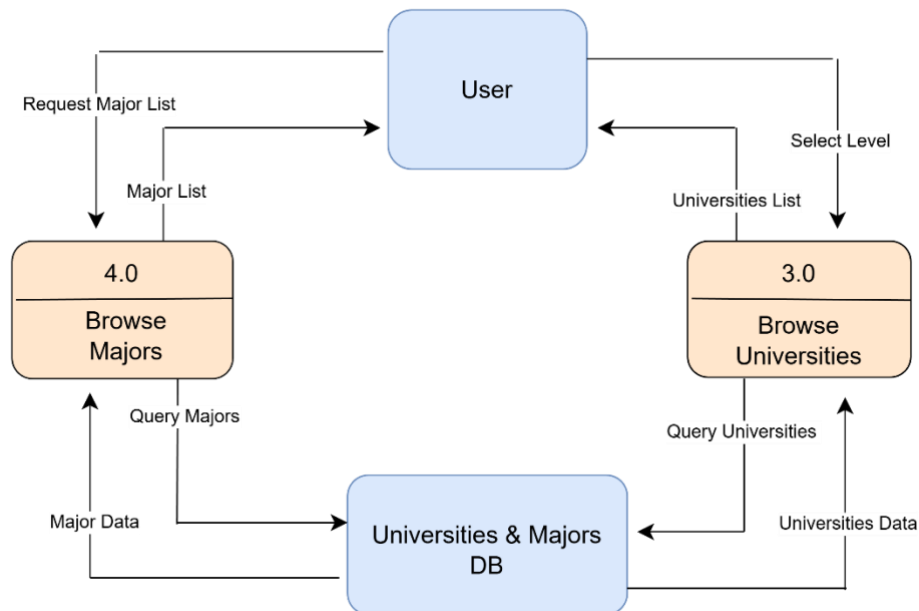


Figure 13 Level-0 DFD of Universities and Majors Subsystem

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6. Data Design

6.1 Data Description

The Ellmuna system deals with two data types: temporary data entered by the user during calculation, and permanent referenced data stored in the database. Temporary data includes high school scores, Qudrat, Tahsily scores, previous GPA, credit hours, and semester course data. This data is processed in memory only and is not stored in the database.

Regarding the permanent data referenced in the system's database to support the "Saudi Universities" feature, this includes degree levels (Bachelor's, Master's, and PhD), universities, colleges, and academic programs. These entities are organized in a hierarchical structure and retrieved when the user browses through the university selection score. Moreover, the application also stores simple settings such as enabling the welcome screen and activating dark mode.

The following tables below describe the main entities, attributes, data types, and constraints:

Entity	Field	Type	PK	FK	Constraint
DegreeLevel	levelID	INT(10)	✓		NOT NULL
	name	VARCHAR(50)			NOT NULL
University	universityID	INT(10)	✓		NOT NULL
	name	VARCHAR(100)			NOT NULL
	city	VARCHAR(50)			NULL
	levelID	INT(10)		✓	NOT NULL
College	collegeID	INT(10)	✓		NOT NULL
	universityID	INT(10)		✓	NOT NULL
	name	VARCHAR(100)			NOT NULL

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Program	programID	INT(10)	✓		NOT NULL
	collegeID	INT(10)		✓	NULL for Bachelor
	universityID	INT(10)		✓	NULL for Master/PhD
	levelID	INT(10)		✓	NOT NULL
	name	VARCHAR(100)			NOT NULL
AppSettings	settingID	INT(10)	✓		NOT NULL
	welcomeEnabled	BOOLEAN			NOT NULL
	darkMode	BOOLEAN			NOT NULL
SemesterCourse (Temp)	creditHours	INT(10)			NOT NULL
	grade	VARCHAR(5)			NOT NULL
HighSchoolInput (Temp)	highschoolScore	FLOAT(5,2)			NOT NULL
	aptitudeScore	FLOAT(5,2)			NOT NULL
	achievementScore	FLOAT(5,2)			NOT NULL
GPAResult (Temp)	semesterGPA	FLOAT(4,2)			NOT NULL
	cumulativeGPA	FLOAT(4,2)			NOT NULL

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	message	VARCHAR(100)			NOT NULL
WeightedResult (Temp)	weightedScore	FLOAT(5,2)			NOT NULL
	category	VARCHAR(50)			NOT NULL

Table 3 main entities, attributes, data types, and constraints

6.2 Data Dictionary

In this section, the entities are listed alphabetically along with their field and description for each:

Entity	Field	Description
AppSettings	welcomeEnabled	Indicates whether the welcome screen is enabled.
	darkMode	Indicates whether dark mode is activated.
College	collegeID	Unique identifier for the college.
	universityID	Identifier linking the college to a university.
	name	Name of the college.
DegreeLevel	levelID	Unique identifier for the academic level.
	name	Name of the academic level.
GPAResult	semesterGPA	Calculated GPA for the current semester.
	cumulativeGPA	Updated cumulative GPA.
	message	Academic status message.
HighSchoolInput	highschoolScore	High school grade and weighting.

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	aptitudeScore	Aptitude test score and weighting.
	achievementScore	Achievement test score and weighting.
Program	programID	Unique identifier for the academic program.
	collegeID	Links program to college (Bachelor).
	universityID	Links program to university (Master/PhD).
	levelID	Academic level of the program.
	name	Name of the academic program.
SemesterCourse	creditHours	Number of credit hours for the course.
	grade	Grade received for the course.
University	universityID	Unique identifier for the university.
	name	Name of the university.
	city	City of the university.
	levelID	Levels the university offers.
WeightedResult	weightedScore	Final computed weighted score.
	category	Classification category.

Table 4 listed entities alphabetically along with their field and description

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6.3 Database Description

The Ellmuna system depends on the relational database, where each entity is stored as a table. This construction supports the application's hierarchical navigation and qualifies a clear conception through ER and schema diagrams

ER diagram:

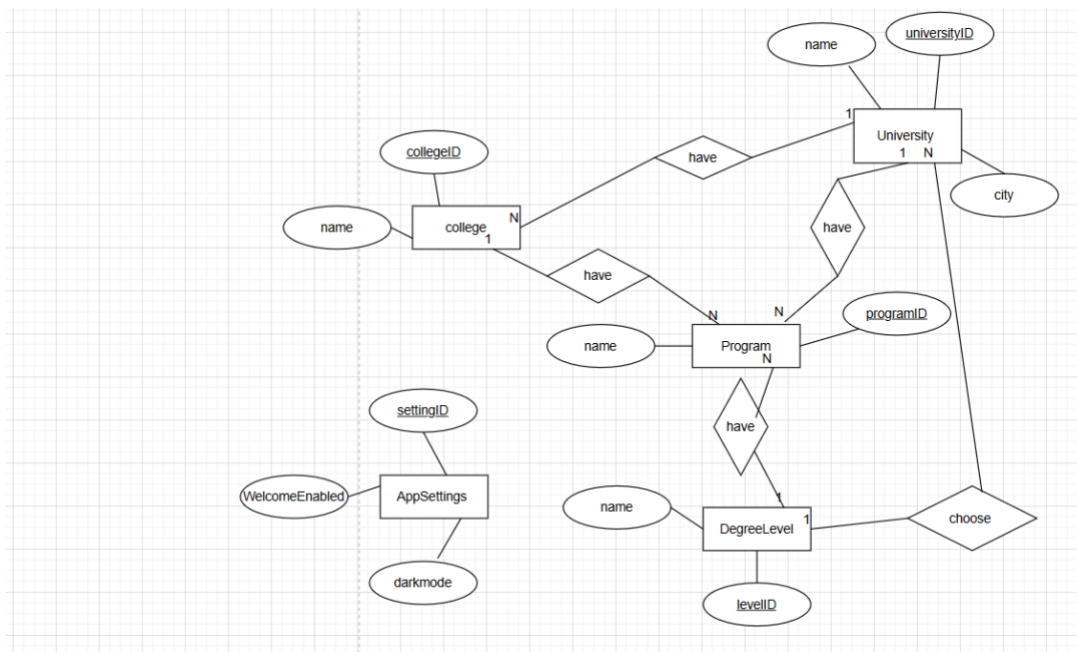


Figure 15 ER digram

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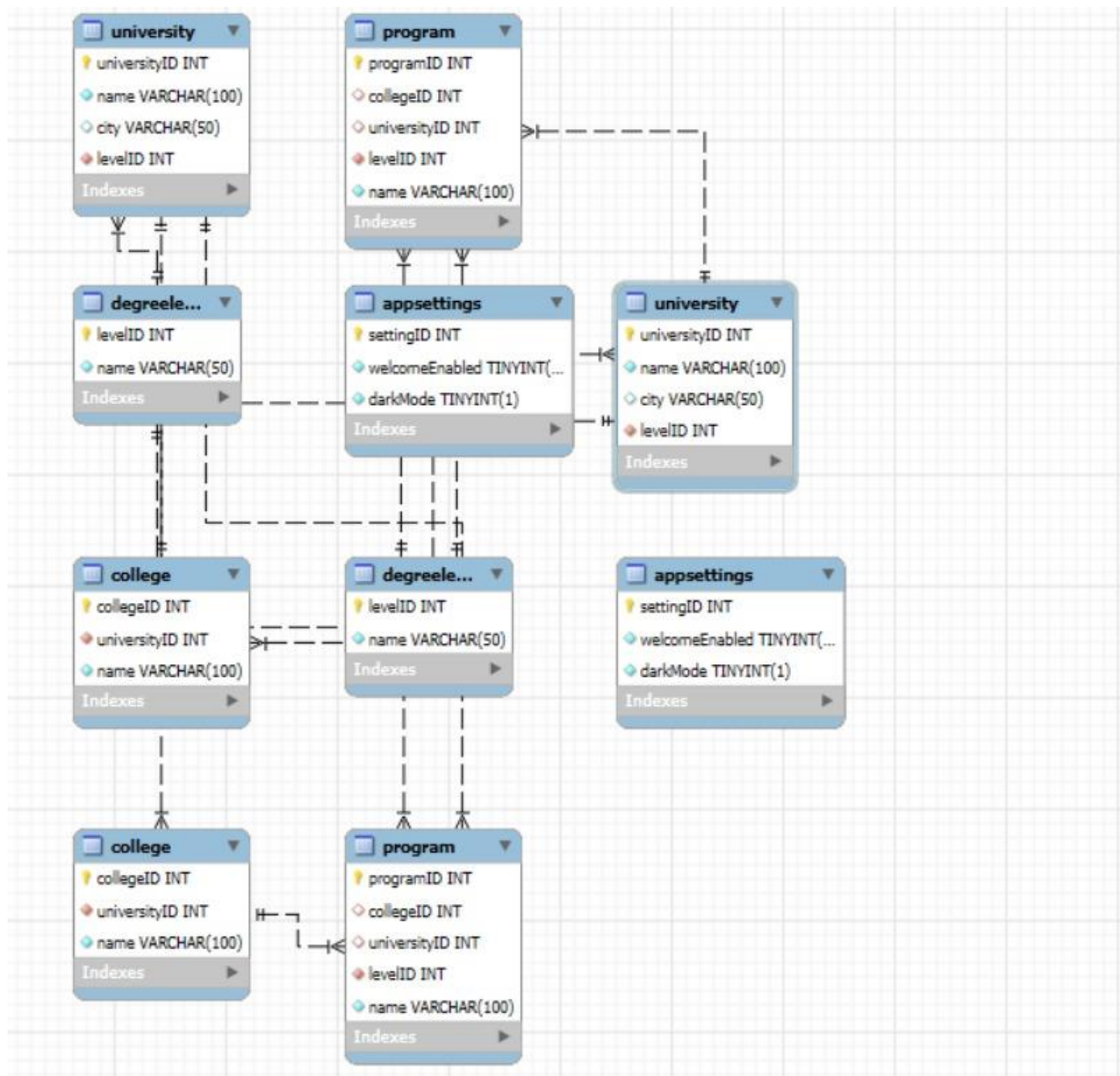


Figure 16 Schema Diagram

7. Component Design

This section presents the pseudocode of the major functionalities of the Ellmuna system in order to give an informal description of the main functions of the system that will utilize the development of the system's algorithms. The pseudocodes expressed below are in natural language.

7.1 Common Functions

7.1.1 Open_Homepage

Open_Homepage(){

 Display Ellmuna logo and main menu options:

- “Weighted Average Calculator”
- “GPA Calculator”
- “Saudi Universities”

 If “Weighted Average Calculator” button is clicked, then

 Redirect to “High School Weighted Average” interface

 If “GPA Calculator” button is clicked, then

 Redirect to “University GPA Calculator” interface

 If “Saudi Universities” button is clicked, then

 Redirect to “Universities – Degree Level” interface}

7.1.2 Open_SideMenu

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Open_SideMenu(){

 If side-menu icon is clicked, then

 Display menu options: “Home”, “Settings”, “Universities”, “Calculator”

 If “Home” is selected, then

 Redirect to “Homepage” interface

 If “Settings” is selected, then

 Redirect to “Settings” interface

 If “Universities” is selected, then

 Redirect to “Universities – Degree Level” interface

 If “Calculator” is selected, then

 Redirect to “Calculator Options” interface}

7.2 Settings Functions

7.2.1 Manage_Settings

Manage_Settings(){

 Display switches: “Enable Welcome Screen”, “Dark Mode”

 If “Enable Welcome Screen” is turned ON, then

 Save preference to show welcome screen on next app launch

 Else

 Save preference to skip welcome screen

 If “Dark Mode” is turned ON, then

 Apply dark theme and save user preference

 Else

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Apply light theme and save user preference}

7.2.2 View_WelcomeScreen

View_WelcomeScreen(){

If “Enable Welcome Screen” preference is ON, then

Display welcome message and description of Ellmuna app

If “Back” button is clicked, then

Redirect to “Homepage” interface}

7.3 High School Weighted Average Functions

7.3.1 Calculate_WeightedAverage_HS

Calculate_WeightedAverage_HS(){

Input High School score

Input Qudrat score

Input Tahsily score

Use default or user-edited weights (e.g., 30%, 30%, 40%)

If “Calculate” button is clicked, then

Compute weighted average percentage

Display result in “Weighted Percentage Result” interface including:

- Final percentage
- Evaluation text (Excellent, Very Good, ...)}

7.3.2 View_WeightedResult_Help

Software Design Specifications for Ellmuna App

View_WeightedResult_Help(){

 If “Grade Scale / Help” button is clicked from result screen, then

 Display grade ranges (Excellent, Very Good, Good, Acceptable, Pass, Fail)

 If “Home Page” button is clicked, then

 Redirect to “Homepage” interface

 If “Back” button is clicked, then

 Return to previous result interface}

7.4 University GPA Functions

7.4.1 Add_Course_Grade

Add_Course_Grade(){

 Input Course Code

 Input Credit Hours

 Input Course Grade (e.g., A+, A, B+, ...)

 Select GPA system (out of 4 or out of 5)

 If “Add Course” button is clicked, then

 Validate inputs

 If inputs are correct, then

 Store course in semester list

 Else

 Display error message “Please enter valid course information”}

7.4.2 Calculate_Semester_GPA

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Calculate_Semester_GPA(){

 If at least one course exists in semester list and “Calculate GPA” is clicked, then

 Convert each letter grade to numeric value according to selected GPA system

 Compute term GPA using $(\text{sum of (grade value} \times \text{credit hours)}) / \text{sum of credit hours}$

 If previous GPA and previous hours are provided, then

 Compute cumulative GPA

 Display Term GPA and Cumulative GPA with evaluation text

Else

 Display message “Please add at least one course to calculate GPA”}

7.4.3 Update_Courses

Update_Courses(){

 If “Update Courses” button is clicked, then

 Allow user to modify or delete any course in the list

 If changes are confirmed, then

 Save updated course list

 Recalculate GPA if requested

 If “Home Page” button is clicked, then

 Redirect to “Homepage” interface}

7.5Universities & Majors Browser Functions

7.5.1 Browse_Universities_By_Degree

Browse_Universities_By_Degree(){

Software Design Specifications for Ellmuna App

Display degree options: “Bachelor’s Degree”, “Master’s Degree”, “PhD Degree”

If a degree level is selected, then

Load and display list of universities offering this degree

Enable search bar and pagination (Previous, page numbers, Next)

If a university card is clicked, then

Redirect to “University Colleges” interface for that university}

7.5.2 Browse_University_Colleges

Browse_University_Colleges(){

For selected university, display its name, city and cards of colleges

If page arrows (Previous / Next) are clicked, then

Load next or previous set of colleges

If a college card is clicked, then

Redirect to “Majors Categories” interface for that college}

7.5.3 Browse_Majors_By_Category

Browse_Majors_By_Category(){

Display categories such as “Science Majors”, “CS & IT Majors”, “Engineering Majors”, etc.

If a category is selected, then

Display list of majors under this category (e.g., Computer Science, Cybersecurity, Software Engineering)

If a major is clicked, then

Display major details (if available) or mark as selected for user reference}

Software Design Specifications for Ellmuna App

Software Design Specifications for Ellmuna App

8. Detailed System Design

Most components described in the System Architecture section will require a more detailed discussion. Other lower-level components and subcomponents may need to be described as well. Each subsection of this section will refer to or contain a detailed description of a system software component. The discussion provided should cover the following software component attributes:

8.1 Classification

In the following sub-section, each component's type, purpose, and responsibilities are described with the help of Table (5). This table presents the classification of all user interface components included in the Elmuna application.

Table 5 Component Classification

Component	Classification	Definition and Responsibility
Home Screen	Interface	This interface displays the application logo and main menu options, allowing the user to navigate to all core system features including weighted average, GPA calculator, universities explorer, and settings.
Weighted Average Calculator Screen (High School Student)	Interface	This interface allows the student to enter high school, Qudrat, and Tahseeli scores, calculate the weighted average, and display a classification message based on the final result.
GPA Calculator Screen (University Student)	Interface	This interface enables the user to select the GPA system (4 or 5), input previous GPA and hours, add semester courses with hours and grades, and calculate both semester and cumulative GPA with a descriptive message.
Universities Explorer Screen	Interface	This interface allows the user to browse Saudi universities by study level (Bachelor, Master, PhD), view available colleges and majors, and search or explore program information easily.
Settings Screen	Interface	This interface provides options for enabling/disabling the welcome screen and toggling between light mode and dark mode.

8.2 Definition

In the following sub section, each component's specific purpose and semantic meaning shall be described with the help of Table (6).

Table 6 Component Definition

Component	Definition
Home Screen	This component defines the main entry point of the application. It introduces the system to the user and provides access to the core features through the primary navigation menu.
Weighted Average Calculator Screen (High School Student)	This component defines the process of entering high school, Qudrat, and Tahseeli scores to compute the final weighted average and deliver a clear result interpretation.

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GPA Calculator Screen (University Student)	This component defines the GPA calculation workflow, enabling users to select their GPA system, enter academic history, add current semester courses, and compute semester and cumulative GPAs.
Universities Explorer Screen	This component defines the university browsing and exploration functionality, allowing users to navigate through study levels, view available colleges and majors, and search for academic programs in Saudi universities.
Settings Screen	This component defines the customization options of the system, enabling users to modify display preferences such as enabling the welcome screen and switching between light and dark modes.

8.3 Responsibilities

in the following sub section, each component's primary responsibilities and expected behavior are described with the help of Table (7). This table outlines what each component accomplishes, the roles it plays within the system, and the services it provides to the user based on the system requirements.

Table 7: Component Responsibilities

Component	Responsibilities
Home Screen	The primary responsibilities of this component are to serve as the main entry point of the system, guide users to the core features, and provide navigation services. It accomplishes the role of an access hub and offers essential routing functionality to its clients.
Weighted Average Calculator Screen (High School Student)	The primary responsibilities of this component are to collect user inputs for high school, Qudrat, and Tahseeli scores, validate the entered data, compute the weighted average based on defined criteria, and display interpretive classification messages. It provides calculation and feedback services.
GPA Calculator Screen (University Student)	The primary responsibilities of this component include enabling users to input academic history, add semester courses, compute semester and cumulative GPA, and show explanatory academic status messages. It provides GPA calculation services and supports users in understanding their academic performance.
Universities Explorer Screen	The primary responsibilities of this component are to allow users to browse universities by level (Bachelor, Master, PhD), view colleges and majors, and search program information. It serves as an exploration and navigation tool for academic programs.
Settings Screen	The primary responsibilities of this component are to allow users to personalize display options, toggle the welcome screen, enable or disable dark mode, and apply changes immediately. It provides customization and preference-management services.

8.4 Constraints

In the following sub section, all relevant assumptions, limitations, and constraints associated with each component are described with the help of Table (8). These constraints include preconditions, postconditions, input/output restrictions, data access rules, synchronization limits, and exception-handling requirements necessary for the correct behavior of each component.

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Table 8 Component Constraints

Component	Constraints
Home Screen	Must load instantly upon opening the app (timing constraint).• Requires stable navigation links to all other components (state constraint).• Must display correctly on different screen sizes (UI format constraint).• User interaction requires no preconditions other than launching the app.
Weighted Average Calculator Screen	Inputs must be numeric and within valid ranges (0–100) — input constraint.• Calculation cannot proceed unless all fields are filled — precondition.• Output classification depends on formula defined in requirements — postcondition.• Must handle invalid input and display validation messages — exception handling.• Data format requires decimal values only.
GPA Calculator Screen	User must select GPA system (4 or 5) before entering data — precondition.• Credit hours must be positive integers — input constraint.• Course grades must match valid grade formats (A+, A, B+, etc.) — data format constraint.• Calculation must follow university GPA rules — invariant.• Cannot compute GPA if required fields are missing — postcondition.• Must handle invalid grade entries — exception constraint.
Universities Explorer Screen	Requires access to stored or fetched university data — data access constraint.• Navigation depends on user's previous selection (Bachelor → University → College → Major) — state constraint.• Must support search without delay — timing constraint.• Must handle cases where some universities or majors are unavailable — exception handling.• Data lists must follow consistent formats.
Settings Screen	Changes must apply immediately to the UI — state constraint.• User preferences must be stored locally — storage constraint.• No operation allowed without user interaction — precondition.• Dark mode switch must update all screens consistently — invariant.• Must handle invalid or unsupported settings gracefully — exception handling.

8.5 Composition

In this subsection, the interactions and collaboration between the system's components are explained through Table (10). The table highlights how each component communicates with other interfaces, what inputs or outputs are exchanged, and what dependencies or side effects may occur during execution. These interactions help clarify the overall system workflow and ensure consistent integration across all subsystems.

Table 9 component composition

Component	Composition
Home Screen	• Application Logo: Represents the system identity and welcomes the user.• Application Name Text: Provides clear branding and recognition for the app.• Main Navigation Buttons: Four buttons that lead to the primary system functions (Weighted Average, GPA Calculator, Universities Explorer, Settings).• Settings Icon: Gives access to user preference configurations.

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Weighted Average Calculator Screen (High School Student)	• High School Score Input Field: Captures the student's secondary percentage. • Qudrat Score Input Field: Allows entering the Qudrat exam score. • Tahseeli Score Input Field: Allows entering the Tahseeli score. • Calculate Button: Performs the weighted average formula using input values. • Result Display Area: Shows the final numerical weighted average. • Classification Message Box: Displays an interpretation message based on the calculated percentage.
GPA Calculator Screen (University Student)	• GPA System Selector: Determines whether the GPA is calculated out of 4 or 5. • Previous GPA Input Field: Stores past cumulative GPA for accurate calculation. • Previous Hours Input Field: Accepts completed credit hours. • Add Course Form: Contains subcomponents for hours and grade input for each course. • Course List Display: Shows all added courses for the current semester. • Calculate GPA Button: Computes semester and cumulative GPA. • Results Display Area: Presents GPA results and academic status messages.
Universities Explorer Screen	• Study Level Selector: Allows choosing (Bachelor, Master, PhD). • Universities List: Displays universities offering the chosen study level. • Colleges List: Shows colleges available within the selected university. • Majors List: Displays majors within the selected college. • Search Bar: Enables searching for universities, colleges, or majors. • Results Display Panel: Shows structured program information.
Settings Screen	• Welcome Screen Toggle: Enables or disables the welcome screen on launch. • Dark Mode Toggle: Switches app appearance between light and dark mode. • Preference Storage: Saves user selections locally so the settings persist.

8.6 Uses/Interactions

In the following subsection, an illustration of each components' interaction and collaboration with other components is provided with the help of Sequence Diagrams.

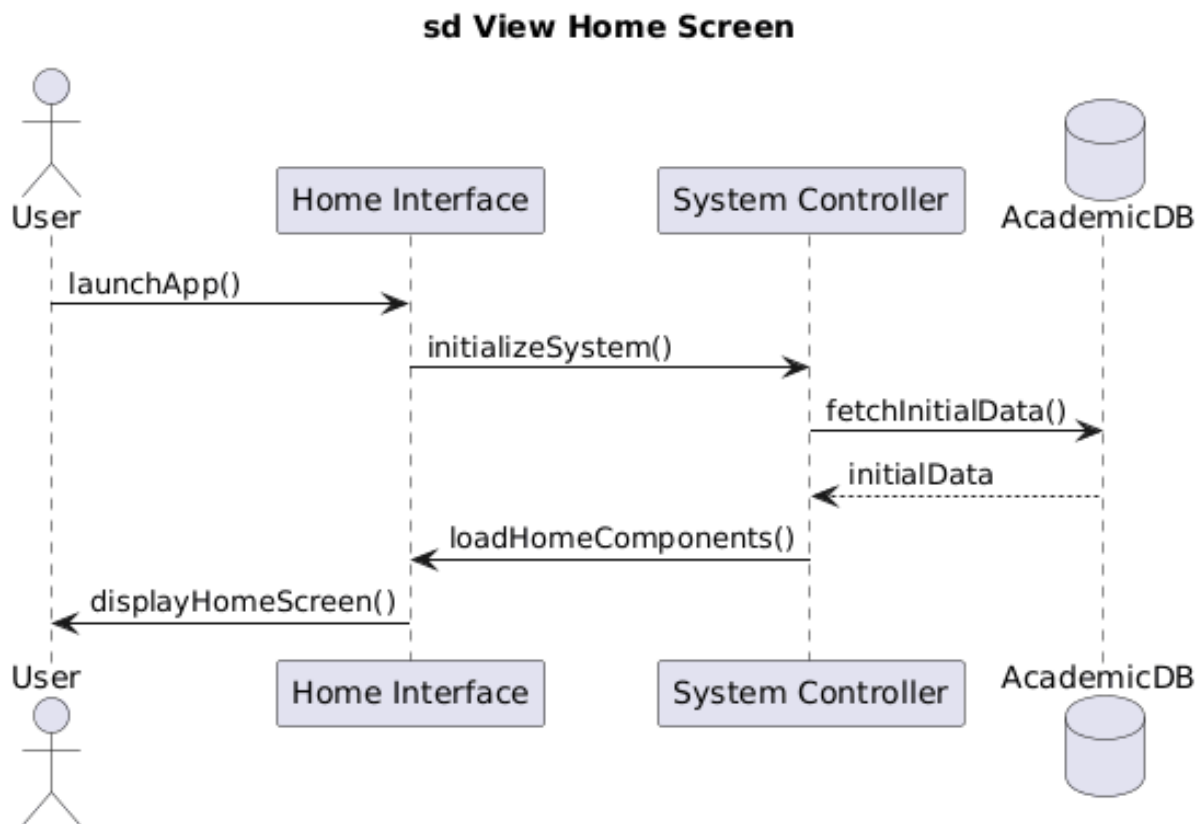


Figure 17 'View Home Screen' Sequence Diagram

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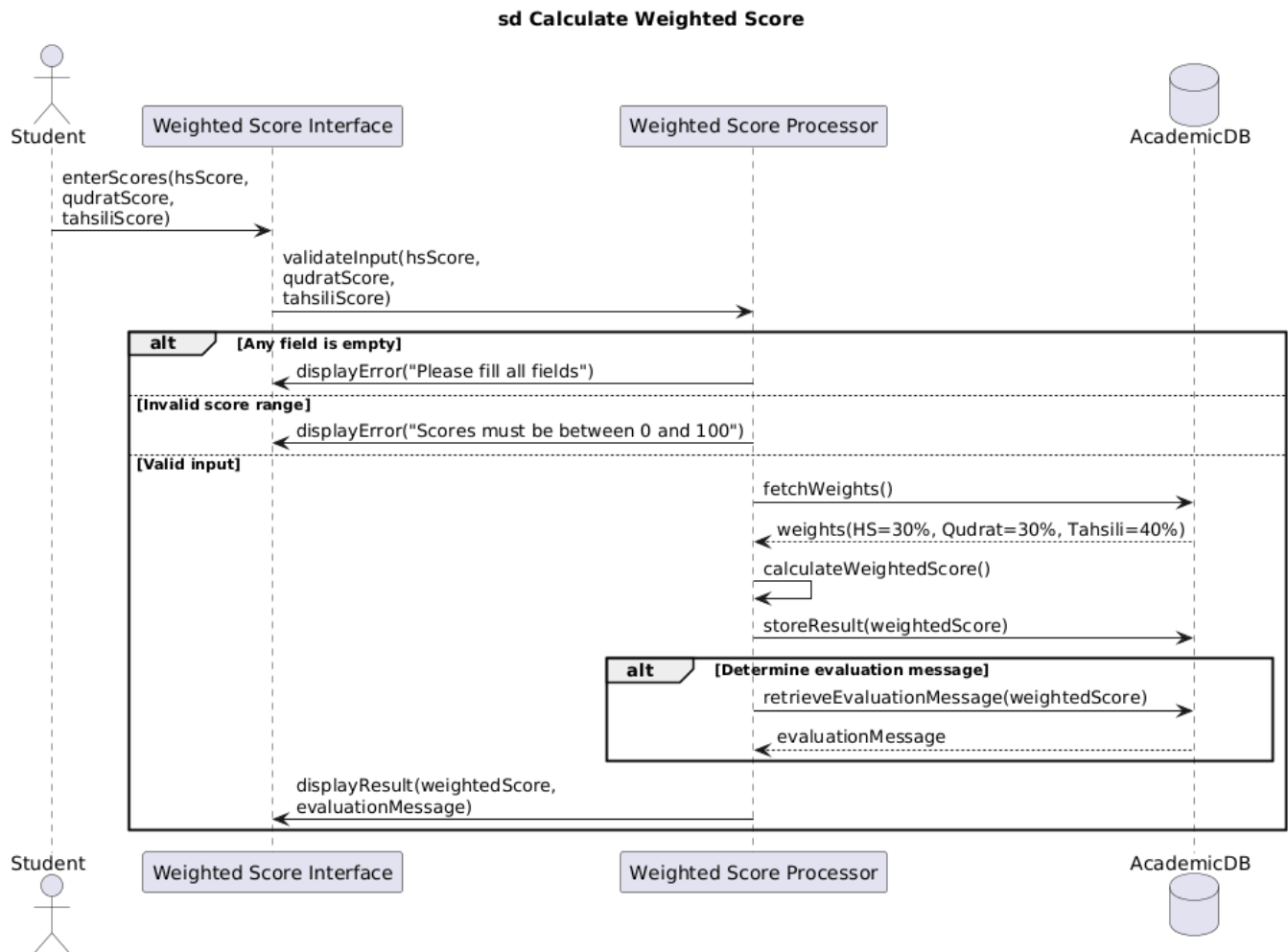


Figure 18 'Calculate Weighted Score' Sequence Diagram

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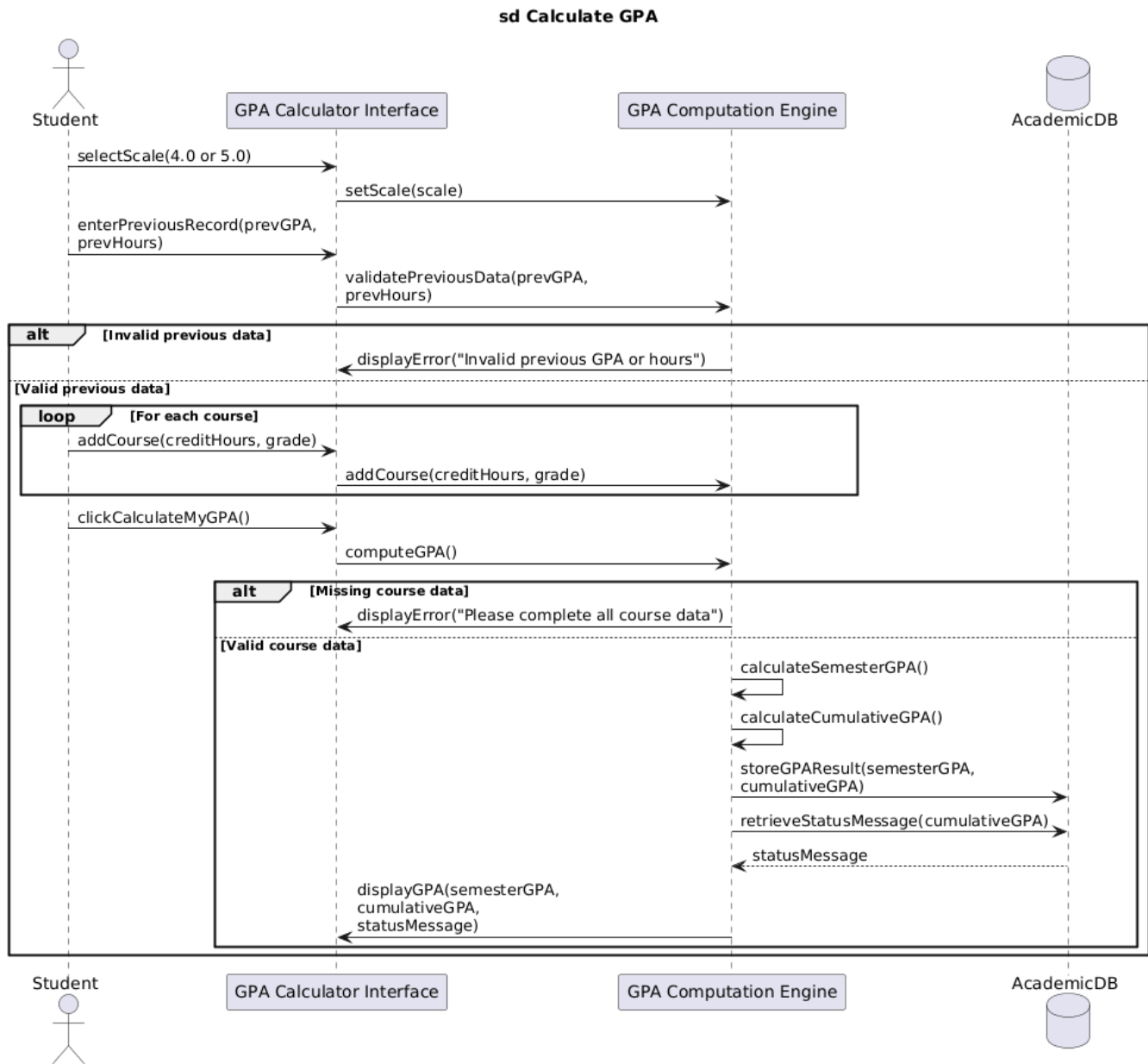


Figure 19 'Calculate GPA' Sequence Diagram

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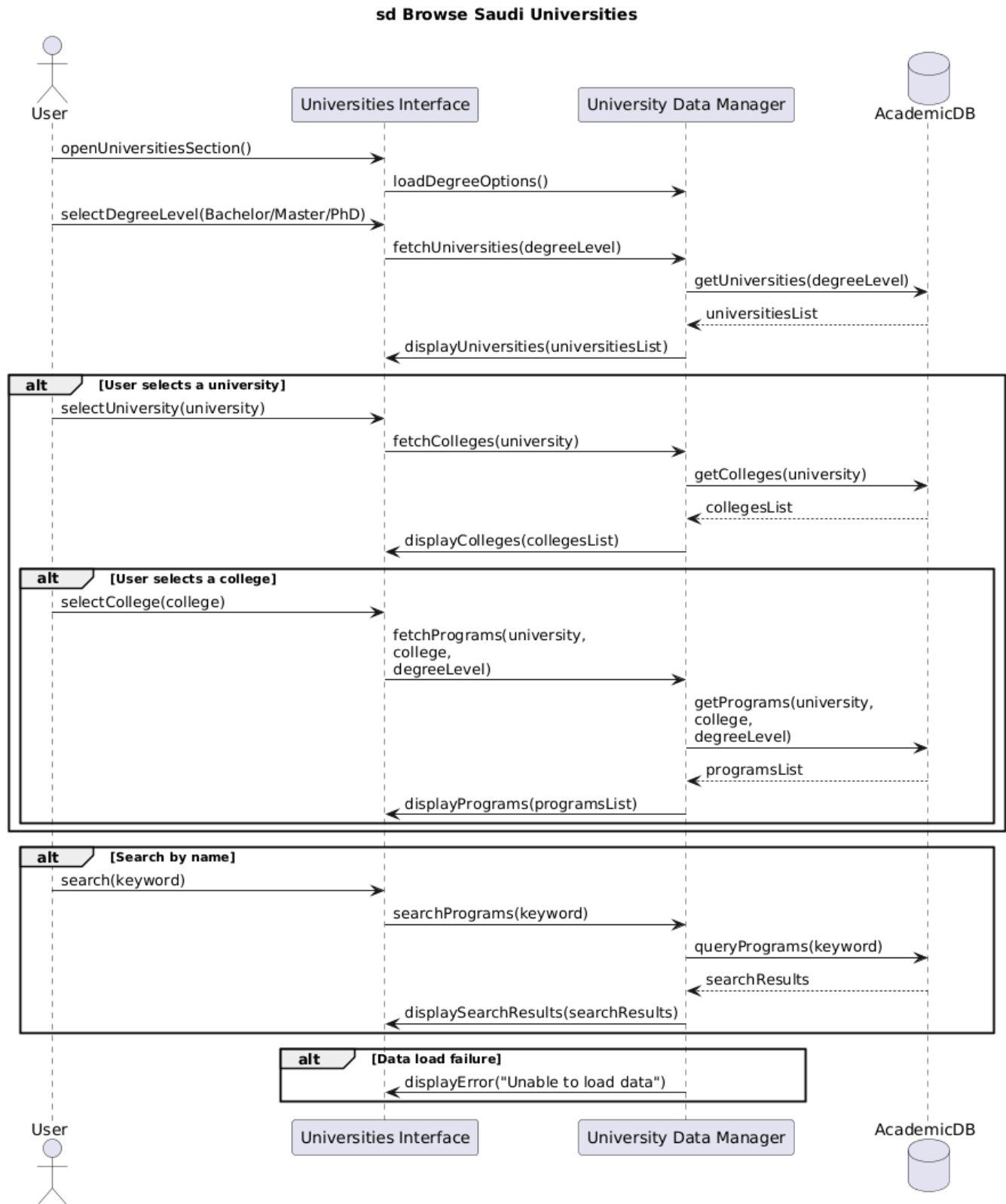


Figure 20 'Browse Saudi Universities' Sequence Diagram

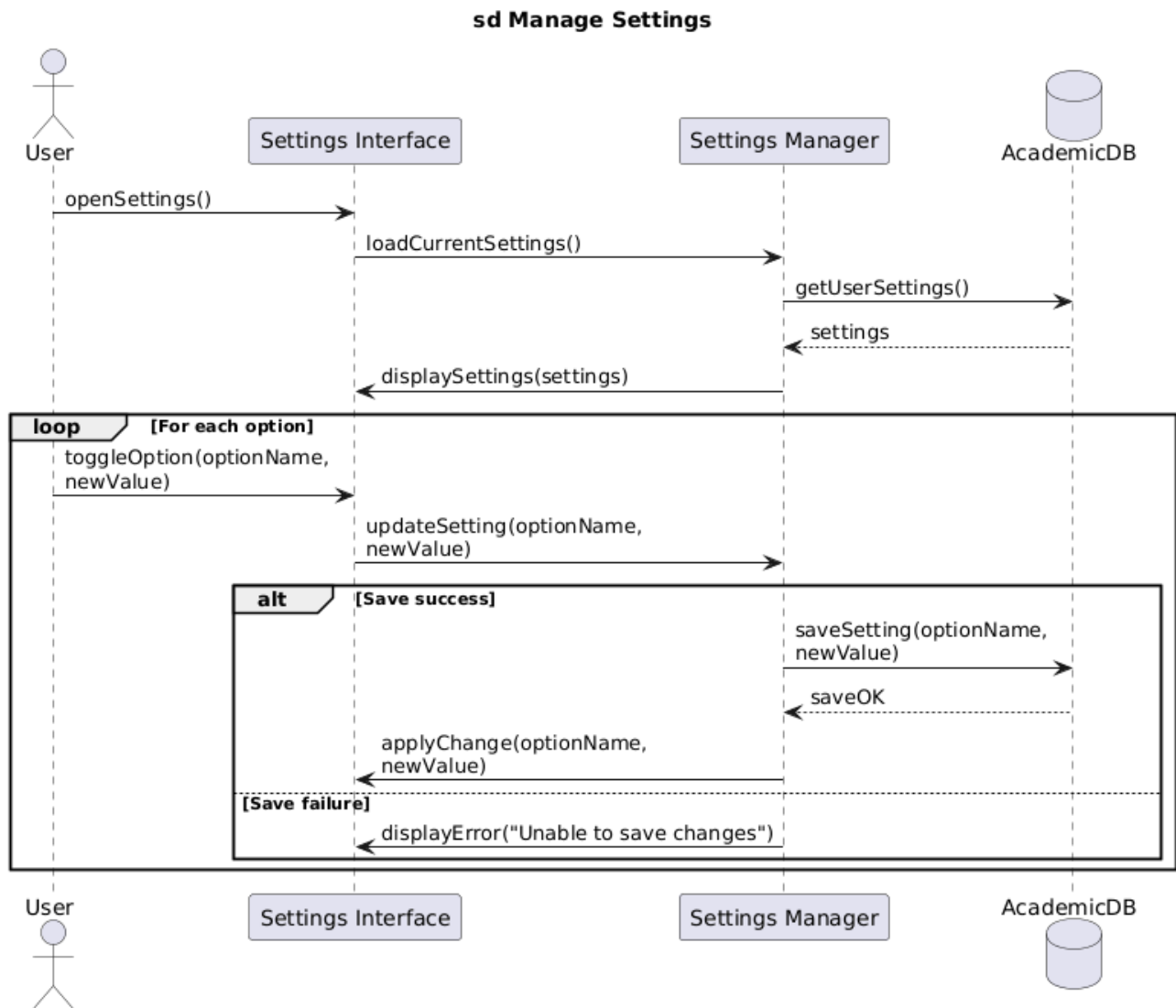


Figure 21 Manage Settings' Sequence Diagram

8.7 Resources

This subsection outlines all resources required by each component, as summarized in Table (11). The listed resources include memory usage, CPU processing needs, UI rendering requirements, external data dependencies, and any storage or network resources necessary for the correct operation of each component.

Table 10 Component Resources

Component	Resources
Home Screen	Basic UI resources only. Requires navigation controller. No additional memory or storage needed.
Weighted Average	Requires memory to hold three numeric inputs. Uses CPU for simple

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Calculator Screen	calculations. Needs UI components for displaying results and classification.
GPA Calculator Screen	Requires temporary memory for course list entries. Uses CPU to process GPA formulas. Needs UI resources for dynamic course display and result areas.
Universities Explorer Screen	Requires access to university, college, and major datasets. Uses memory for displaying hierarchical lists. Requires search resources and possibly network resources if data is external.
Settings Screen	Requires local storage to keep user preferences. Uses minimal CPU and UI rendering resources for theme switching.

8.8 Processing

This subsection describes how each component performs its internal processing tasks, as detailed in Table (12). The processing activities include data validation, formula execution, UI updates, retrieval of academic records, preference handling, and any internal logic needed to deliver the component's intended functionality.

Table 11 Component Processing

Component	Processing
Home Screen	Loads UI elements and waits for user input. Processes only navigation actions.
Weighted Average Calculator Screen	Validates numeric inputs. Applies weighted average formula. Generates classification based on final result. Updates UI with final output.
GPA Calculator Screen	Validates GPA system selection and data input. Processes each course (grade → numeric value → weighted points). Calculates semester and cumulative GPA. Displays calculated values and academic status.
Universities Explorer Screen	Loads academic data. Filters lists based on selected level. Updates university → college → major navigation. Performs search queries and handles empty results.
Settings Screen	Reads/writes preferences to local storage. Applies UI theme changes instantly. Updates related screens affected by dark/light mode.

8.9 Interface/Exports

This section describes the set of services, resources, data elements, constants, and exceptions that are exported by each component. These exported elements represent what the component provides to the rest of the system, including any values, results, configurations, or interaction features. Each export includes its meaning, usage context, and any rules or constraints associated with it.

For each component, the exported elements reflect the functional behavior defined in earlier sections (Classification, Definition, Responsibilities, Constraints, Composition, Uses, Resources, and Processing).

Home Screen — Interface / Exports

- Navigation Exports: Provides routing access to the Weighted Average Calculator, GPA Calculator, Universities Explorer, and Settings components.
- UI Labels and Actions: Exports the main menu buttons used by other components to trigger navigation.
- State Export: Indicates which component was selected by the user to initiate the next screen.

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Weighted Average Calculator Screen — Interface / Exports

- Computed Weighted Average: Exports the final numeric weighted average result.
- Classification Message: Exports an interpretation message based on predefined percentage ranges.
- Input Validation Errors: Exports exception messages when invalid scores or missing fields occur.
- Score Data Types: Exports numeric values (0–100) as input constraints.
- Calculation Rule Constant: Exports the fixed formula used for computing the weighted average.

GPA Calculator Screen — Interface / Exports

- Semester GPA Result: Exports the computed semester GPA.
- Cumulative GPA Result: Exports the final cumulative GPA after processing new semester data.
- Academic Status Message: Exports indicators such as honors, warning, or satisfactory standing.
- Course Data Structures: Exports the list of added courses (credit hours + grade).
- Validation Exceptions: Exports error messages for invalid GPA entries or unsupported grade formats.

Universities Explorer Screen — Interface / Exports

- Universities List: Exports the list of universities available for the selected study level.
- Colleges List: Exports the colleges belonging to a selected university.
- Majors List: Exports the academic majors available in the selected college.
- Search Results: Exports filtered results for universities, colleges, and majors.
- Empty Result Exceptions: Exports a message when the search yields no results.
- Structured Data Types: Exports hierarchical academic data in list format.

Settings Screen — Interface / Exports

- Theme Mode State: Exports the current UI mode (Light or Dark).
- Welcome Screen Preference: Exports whether the welcome screen is enabled or disabled.
- Updated UI Rules: Exports appearance changes that affect all other components.
- Stored Preferences: Exports persistent settings saved in local storage.
- Invalid Setting Exceptions: Exports messages for unsupported configuration attempts.

8.10 Detailed Subsystem Design

This section presents an in-depth description of the subsystems that constitute the “Elmuna App.” Each subsystem groups together several related components that collectively fulfill a major functional objective. The following description outlines each subsystem’s purpose, internal structure, behavior, information/control flow, and interactions with other subsystems.

- Calculation Subsystem

Included Components

- Weighted Average Calculator Screen
- GPA Calculator Screen

Subsystem Purpose

The Calculation Subsystem is responsible for all academic computations within the application. It processes user inputs, validates data, applies predefined formulas, and generates accurate numerical results accompanied by classification or status messages.

Internal Structure and Behavior

- The subsystem uses validation rules to ensure inputs fall within acceptable constraints before computation.
- The Weighted Average Calculator applies the standardized weighting model based on high-school, Qudrat, and Tahsili score distributions.
- The GPA Calculator processes academic records—including previous GPA, completed credit hours, and semester courses—to compute both semester and cumulative GPA using official academic formulas.
- Outputs are formatted for clarity and immediately rendered through the corresponding UI.

Information and Control Flow

1. User enters academic data in the corresponding calculator interface.
2. The subsystem verifies data accuracy and completeness.
3. Calculation logic executes using the required formula.
4. Results and classification messages are returned to the UI layer.

Subsystem Interactions

- Interacts with the User Interface & Navigation Subsystem to present results.
- Incorporates visual settings from the Settings Subsystem (e.g., dark mode).

- Academic Exploration Subsystem

Included Components

Software Design Specifications for Ellmuna App

- Universities Explorer Screen

Subsystem Purpose

This subsystem enables structured exploration of academic institutions and programs in Saudi Arabia. It allows browsing by study level and provides detailed access to universities, colleges, and majors.

Internal Structure and Behavior

- Uses hierarchical datasets structured by level → university → college → major.
- Implements filtering and search logic to help users quickly locate institutions or programs.
- Dynamically updates the displayed results based on navigation selections or search queries.

Information and Control Flow

- User selects a study level (Bachelor, Master, PhD).
- Subsystem loads the corresponding list of universities.
- Upon university selection, available colleges are displayed.
- Upon college selection, the related majors appear.
- Optional search queries filter displayed data.

Subsystem Interactions

- Works with the UI subsystem for rendering hierarchical lists.
- May depend on local or external data sources for updated academic information.

9 Other Design Features

The Ellmuna application features a diverse range of additional design elements that are not mentioned in the previous sections. The application provides interface customization preferences, containing the capability to activate or deactivate the welcome screen and switch between bright and dark mode, enabling users to tailor the interface to their preferences. All calculation operations, including GPA computation, weighted-score processing, and semester course handling, are carried out in memory without storing any personal data, which increases privacy and reduces storage requirements. In addition, the system uses a clear hierarchical navigation structure (Degree Level – University - college - program), which makes it possible for users to rapidly and neatly obtain academic material.

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10 Requirements Traceability Matrix

Association ID in SRS	Technical Assumptions / Customer Needs	Functional Requirement	User	Association ID in SDS	System Component
REQ-1	Valid device, app installed	Display Home Screen	General User	7.1	Homepage UI
REQ-2	Stable UI rendering	Show logo, name, buttons	General User	7.1	Homepage UI
REQ-3	Navigation enabled	Correct redirection	General User	7.1	Homepage UI
REQ-4	Usable interface	Simple user-friendly UI	General User	7.1	Homepage UI
REQ-5	Valid HS inputs	Input three academic scores	High School Student	7.3	Weighted Calculator UI
REQ-6	Default values loaded	Use default weights	High School Student	7.3	Weighted Calculator UI
REQ-7	Correct formula	Calculate weighted score	High School Student	7.3	Weighted Calculator Logic
REQ-8	Result evaluation	Display evaluation message	High School Student	7.3	Weighted Result UI
REQ-9	Input validation	Handle invalid input	High School Student	7.3	Error Handling Module
REQ-10	Select scale available	Select GPA scale	University Student	7.4	GPA Scale UI
REQ-11	Correct GPA formula	Calculate GPA accurately	University Student	7.4	GPA Logic
REQ-12	Clear results	Display GPA results	University Student	7.4	GPA Result UI
REQ-13	Validation	Handle invalid course data	University Student	7.4	Error Handling Module
REQ-14	Temp session	Maintain recalculation session	University Student	7.4	Session Manager
REQ-15	Categorized data	Categorize universities	General User	7.5	Universities Browser UI
REQ-16	Updated DB	Display programs/hierarchy	General User	7.5	Universities DB View
REQ-17	Search function	Provide search	General User	7.5	Search Module

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REQ-18	Accurate data	Display accurate data	General User	7.5	Data Module
REQ-19	Smooth navigation	Smooth transitions	General User	7.5	Navigation Controller
REQ-20	Preference storage	Toggle welcome screen	General User	7.2	Settings UI
REQ-21	Theme support	Toggle dark/light mode	General User	7.2	Theme Controller
REQ-22	Instant apply	Apply changes instantly	General User	7.2	Settings Logic
REQ-23	Save prefs	Save user preferences	General User	7.2	Preferences Storage

Table 12 Requirements Traceability Matrix